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RESEARCH TRACK

Word-Level American Sign Language Recognition using Hand Landmarks and Attention-based Recurrent Networks

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Abstract

Sign language recognition is an important accessibility challenge, it serves to mitigate communication barriers between deaf or hard-of-hearing signers and non-signing people. This Final Degree Project develops and evaluates a research prototype for word-level American Sign Language recognition from short video clips. The final approach extracts 21 three-dimensional hand landmarks per frame with MediaPipe Hands, normalises the resulting sequences, and classifies ASL glosses using a bidirectional GRU with a Temporal Attention mechanism. The work also documents the methodological evolution from YOLO-based hand detection and static letter recognition toward landmark-based temporal modelling, where the main challenge lies in representing sign motion rather than localising hands. The system was validated through experiments on WLASL and MS-ASL, including architecture comparison, sequence-length analysis, cross-dataset evaluation, and vocabulary scaling. Results indicate promising performance for reduced vocabularies and reproducible behaviour across datasets, but also clear limitations as the number of classes increases. The landmark representation provides a privacy-friendly abstraction, although the prototype remains a research system rather than a production-ready translator.

Keywords: Sign Language Recognition, American Sign Language, Hand Landmarks, MediaPipe, GRU, Temporal Attention

Resumen

El reconocimiento automático de lengua de signos constituye un reto relevante para mejorar la accesibilidad y reducir barreras de comunicación entre personas sordas o con discapacidad auditiva y usuarios que no conocen la lengua de signos. Este Proyecto Fin de Grado desarrolla y evalúa un prototipo de investigación para el reconocimiento de palabras en American Sign Language a partir de vídeos breves. La solución final utiliza MediaPipe Hands para extraer 21 puntos tridimensionales de la mano por fotograma, normaliza las secuencias obtenidas y las clasifica mediante una red GRU bidireccional con un mecanismo de Atención Temporal. El trabajo recoge también la evolución metodológica desde una primera fase basada en detección de manos con YOLO y reconocimiento estático de letras hacia un enfoque centrado en la representación temporal del signo. La validación se realiza sobre WLASL y MS-ASL, incluyendo comparación de modelos, análisis de longitud de secuencia, evaluación cruzada y escalado de vocabulario. Los resultados son prometedores en vocabularios reducidos, aunque muestran limitaciones al aumentar el número de clases. La representación mediante landmarks aporta ventajas de privacidad, pero el sistema debe entenderse como prototipo experimental, no como traductor listo para producción.

Palabras clave: reconocimiento de lengua de signos, American Sign Language, landmarks de mano, MediaPipe, GRU, atención temporal

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Chapter 1

Introduction

1.1 Problem description

Sign language recognition is located at the intersection of accessibility, computer vision, and sequence modelling. Sign languages are natural languages with their own visual-spatial structure, not simple gesture codes or direct encodings of spoken languages. Their meaning is expressed through hand shape, movement, orientation, location, timing, and, in many cases, non-manual cues such as facial expression or body posture. For this reason, automatic recognition from video is a demanding problem: the same sign may vary across signers, camera viewpoints, execution speeds, lighting conditions, and recording devices.

This Final Degree Project studies American Sign Language (ASL) recognition as an accessibility-oriented research problem. The objective is not to replace professional interpreters or to deliver a production-ready translator, but to evaluate whether a compact visual representation combined with temporal neural modelling can support feasible isolated-word recognition under reproducible experimental conditions. Isolated word-level recognition is therefore treated as a realistic intermediate step between static alphabet classification and full continuous sign language translation. It preserves the temporal nature of signs while keeping the scope appropriate for a research-oriented undergraduate project.

The central research question of the thesis is whether hand-landmark sequences can provide enough information for a temporal model to classify short ASL video clips into word glosses. The final pipeline uses MediaPipe Hands to extract 21 three-dimensional hand landmarks per detected hand and frame [1]. These landmark sequences are normalised and then processed by a bidirectional GRU with a Temporal Attention mechanism. This design reduces the dimensionality of the video input, limits the dependence on background appearance and clothing, and offers a more privacy-preserving representation than storing or processing full RGB frames. At the same time, the approach has clear limitations: it depends on the reliability of hand detection and does not capture all linguistic information present in non-manual markers.

The technical direction of the project evolved during development. Initial experiments explored YOLO-based hand detection and static ASL letter recognition. These experiments were useful for understanding the visual recognition pipeline, but they also showed that hand localisation and static

classification did not address the main difficulty of word-level signing: temporal representation. The project therefore pivoted towards landmark-based sequence modelling, with the experimental work centred on public ASL datasets, mainly WLASL [2] and MS-ASL [3].

The thesis is situated between research and engineering. It includes data preparation, model comparison, sequence-length analysis, WLASL validation, MS-ASL50 cross-dataset validation, vocabulary scaling, limitations analysis, and an ethical and social-impact discussion. A mobile and server-based runtime prototype is also developed as a proof of concept, but the main scientific contribution is the experimental evaluation of the landmark-based recognition approach rather than the deployment of an assistive product.

1.2 Outline

The remainder of the thesis is organised as follows:

- Chapter 2 presents the State of the Art, covering the computer vision and deep learning background required for the project, including convolutional neural networks, pose estimation, MediaPipe Hands, recurrent networks, GRUs, attention mechanisms, and previous work in sign language recognition.
- Chapter 3 defines the main objective, the specific objectives, and the validation methods used to determine whether each objective has been achieved.
- Chapter 4 describes the methodology and work plan. It explains the iterative experiment-driven process, the main project phases, the methodological pivot from static letter recognition to landmark-based word recognition, and the resources and constraints that shaped the project.
- Chapter 5 presents the Technical Development of the system. It details the data preparation pipeline, the training and evaluation pipeline, the selected model architecture, the reproducibility mechanisms, and the runtime proof of concept based on a mobile client and FastAPI server.
- Chapter 6 reports the experimental evaluation. It includes architecture comparison, sequence-length analysis, WLASL30 validation, MS-ASL50 cross-dataset validation, vocabulary scaling, qualitative validation design, and the main limitations observed during experimentation.
- Chapter 7 analyses the ethical implications and social impact of the project, with attention to accessibility, privacy, bias, reproducibility, responsible deployment, and the social value of assistive technologies.
- Chapter 8 reflects on the academic journey at Universidad Francisco de Vitoria and relates the project to the competences developed during the degree.
- Chapter 9 presents the conclusions and future work, summarising the main findings, the limitations of the current prototype, and possible directions for extending the research.
- The bibliography lists the academic and technical sources used throughout the thesis.

Chapter 2

State of the Art

2.1 From Symbolic to Subsymbolic AI

Early Artificial Intelligence (AI) systems were symbolic: they represented knowledge using explicit rules and logic, and solved problems through handcrafted if/then decision trees. This paradigm achieved success in constrained domains, but it struggled with perception tasks such as vision and speech because the real world is noisy, ambiguous, and high-dimensional. The so-called "knowledge acquisition bottleneck" meant that as tasks grew in complexity, engineering and maintaining rule bases became impossible. In the context of sign language, symbolic systems could encode simple gesture prototypes but failed to capture the fine-grained variability across signers, viewpoints, and execution speeds.

The shift to subsymbolic AI reframed intelligence as pattern recognition. Instead of encoding expert rules, models learn distributed representations from data. This change was prompted by advances in statistical learning and neural networks, culminating in deep learning methods that learn hierarchical features directly from raw inputs [4]. Subsymbolic models are particularly effective in perception because they optimize representations for the task objective, allowing subtle visual patterns to emerge without explicit programming. In sign language recognition, subsymbolic models learn the visual semantics of hand shapes, motions, and temporal dynamics from examples rather than hand-coded descriptors.

From a theoretical standpoint, subsymbolic approaches replace symbolic inference with continuous function approximation. A deep network implements a parameterized mapping f_θ that transforms inputs into task-specific embeddings, optimized by minimizing a loss over data. This paradigm aligns with the "computational" and "algorithmic" levels of explanation in vision, where representation learning is the core challenge. By training end-to-end models on large datasets, modern systems discover invariances to lighting, background, and signer variability that are nearly impossible to enumerate symbolically.

The subsymbolic shift also enabled modular pipelines in which perceptual frontends (e.g., hand pose estimation) feed compact, structured features to temporal models. This decoupling is powerful for SLR: the vision module learns spatial representations, while the sequence model learns dynamics. The result is a scalable, data-driven framework that has largely replaced rule-based approaches in

the literature. Contemporary SLR therefore reflects the broader AI trajectory: from symbolic rules to learned representations and, increasingly, to architectures that encode inductive biases (e.g., convolutional and recurrent structures) while remaining fully trainable from data.

2.2 Feature Extraction and Pose Estimation

The first challenge in Sign Language Recognition is extracting meaningful spatial features from video frames. This field has evolved from processing raw pixels to extracting high-level geometric structures.

2.2.1 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) represent the foundational technology for modern computer vision [4]. Unlike fully connected networks that treat inputs as flat vectors, CNNs preserve the spatial structure of images through three main operations: convolution, non-linearity, and pooling. A typical convolutional layer computes a feature map $\mathbf{y}$ as:

$$\mathbf{y}_{i,j,k} = \sigma\left(\sum_{c=1}^{C}\sum_{u,v}\mathbf{w}_{u,v,c,k}\,\mathbf{x}_{i+u,j+v,c} + b_k\right) \tag{2.1}$$

where $\mathbf{x}$ is the input tensor, $\mathbf{w}$ are learned kernels (filters), and σ is a non-linearity (e.g., ReLU). This operation introduces an inductive bias toward *translation equivariance*, meaning the network can recognize features regardless of their position in the image.

Deep CNN architectures, such as AlexNet [5] or ResNet, stack many such layers to learn hierarchical representations. Initial layers typically behave as Gabor filters, detecting simple edges and textures. Deeper layers combine these low-level features to detect complex shapes, and finally, semantic objects like hands or faces. The use of pooling layers further enhances robustness by introducing local translation invariance, allowing the network to detect features even if they are slightly shifted or distorted.

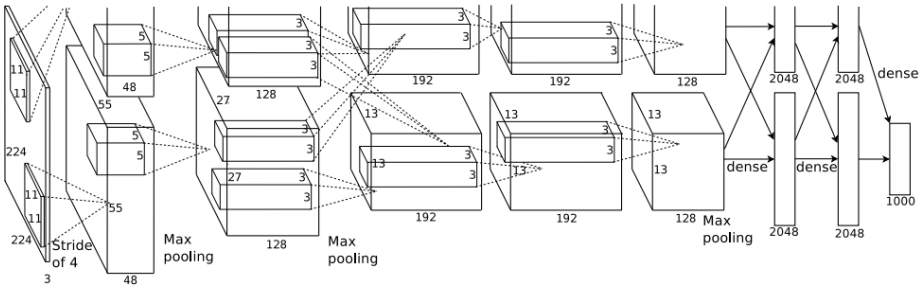


Figure 2.1: Architecture of AlexNet. Note the split processing paths originally designed for dual-GPU training. Source: Krizhevsky et al. [5].

2.2.2 From Pixels to Landmarks: Pose Estimation

While CNNs can process raw RGB pixels directly (pixel-based approach), this method is computationally expensive and sensitive to background noise and lighting conditions. To address this, *Pose Estimation* techniques leverage CNNs to regress the coordinates of anatomical keypoints, transforming the visual problem into a geometric one [6]. In practice, pose pipelines are commonly framed as either top-down (detect hands first, then regress keypoints) or bottom-up (predict keypoints first, then group them into hands). For SLR, the top-down approach is especially attractive because it focuses computation on the hand region and naturally supports frame-to-frame tracking.

MediaPipe Hands [1] is a seminal system in this domain. It employs a two-stage pipeline to achieve real-time performance:

1. **Palm Detector:** A lightweight single-shot detector operates on the full image to locate the hand palm. This step is crucial because detecting a rigid object like a palm is simpler than detecting articulating fingers.
2. **Landmark Regressor:** A high-capacity CNN processes the cropped hand region to predict the 3D coordinates of 21 skeletal joints.

This architecture decouples detection from regression, allowing the system to track hands efficiently by re-using the crop from the previous frame. For SLR, this provides a compact, invariant representation of the hand shape, filtering out irrelevant visual information such as background clutter or skin tone variations. The landmark representation also enables geometric normalization (e.g., translation and scale alignment) and the derivation of higher-level features such as bone vectors, joint angles, and inter-finger distances, which often yield more stable temporal dynamics than raw pixels.

Pose-based pipelines frequently integrate additional cues beyond the hands. Upper-body or face landmarks can capture non-manual signals (head pose, facial expression) that disambiguate signs or encode grammatical structure. Even when only hand keypoints are used, modern systems typically propagate per-joint confidence scores to handle occlusions and motion blur, masking unreliable points rather than forcing the sequence model to fit noise. These design choices make the pose representation more robust to real-world variability in signer appearance, speed, and viewpoint.

2.3 Sequence Modeling

Since sign language is inherently temporal, recognizing signs requires modeling the evolution of features over time. Standard feedforward networks cannot handle this task because they accept fixed-size inputs and have no memory of previous states. This limitation necessitates the use of Recurrent Neural Networks (RNNs).

2.3.1 Recurrent Neural Networks (RNNs)

RNNs introduced the concept of "memory" in neural networks by maintaining a hidden state $\mathbf{h}_t$ that is updated at each time step based on the current input $\mathbf{x}_t$ and the previous state $\mathbf{h}_{t-1}$:

$$\mathbf{h}_t = \phi(\mathbf{W}_{xh}\mathbf{x}_t + \mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{b}_h) \tag{2.2}$$

This recurrent connection allows the network to process sequences of variable length, sharing the same parameters (weights) across all time steps. Training is performed using *Backpropagation Through Time* (BPTT), which unrolls the network graph for the duration of the sequence.

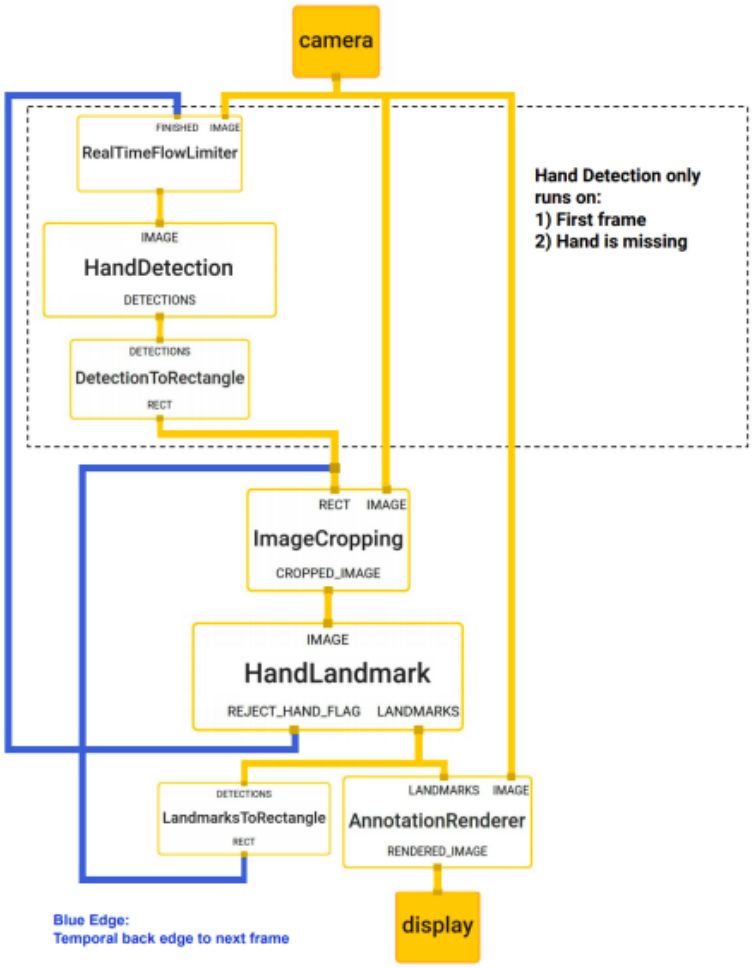


Figure 2.2: MediaPipe Hands pipeline showing the interaction between the detection and landmark models. Source: Zhang et al. [1].

However, RNNs suffer from a critical flaw known as the *vanishing gradient problem*. During BPTT, gradients are calculated by the chain rule, which involves repeated multiplication of the weight matrix. If the eigenvalues of this matrix are smaller than 1, the gradients decay exponentially as they propagate back to earlier time steps. Consequently, the network fails to learn long-term dependencies, effectively "forgetting" the beginning of a sign by the time it reaches the end.

2.3.2 Long Short-Term Memory (LSTM)

To overcome the vanishing gradient problem, Hochreiter and Schmidhuber [7] proposed the Long Short-Term Memory (LSTM) network. The core innovation of the LSTM is the separation of the

memory into two distinct vectors: the *Hidden State* ($\mathbf{h}_t$), which carries short-term information, and the *Cell State* ($\mathbf{C}_t$), which acts as a "highway" for information flow.

The Cell State runs down the entire chain with only minor linear interactions, making it easy for information to flow unchanged. This flow is regulated by three distinct "gates" composed of sigmoid neural network layers:

- **Forget Gate:** Decides what information to discard from the cell state (outputs numbers between 0 and 1).
- **Input Gate:** Decides what new values will be updated in the cell state.
- **Output Gate:** Controls what parts of the cell state are output to the hidden state.

This mechanism, known as the *Constant Error Carousel* (CEC), protects the error signal from decaying, enabling LSTMs to bridge time lags of more than 1000 steps. While highly effective, the complexity of calculating four distinct linear layers for each time step results in high computational costs and parameter counts. The internal structure of an LSTM cell is shown in **Figure 2.3(a)**.

2.3.3 Gated Recurrent Units (GRU)

Gated Recurrent Units (GRUs), proposed by Chung et al. [8], represent a modern simplification of the LSTM architecture. GRUs address the redundancy in LSTMs by removing the separate cell state and merging the input and forget gates into a single **Update Gate** ($\mathbf{z}_t$). They also introduce a **Reset Gate** ($\mathbf{r}_t$).

The mathematical formulation for a GRU at time t is:

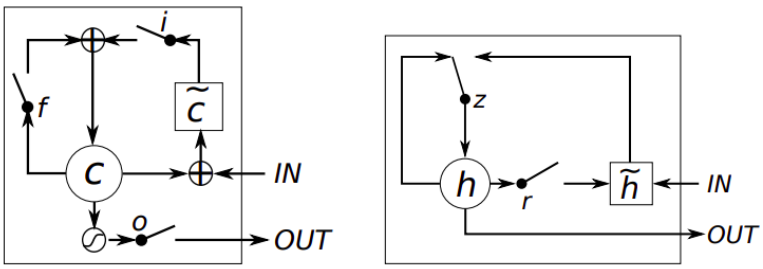
$$\mathbf{z}_t = \sigma(\mathbf{W}_z \mathbf{x}_t + \mathbf{U}_z \mathbf{h}_{t-1} + \mathbf{b}_z) \tag{2.3}$$

$$\mathbf{r}_t = \sigma(\mathbf{W}_r \mathbf{x}_t + \mathbf{U}_r \mathbf{h}_{t-1} + \mathbf{b}_r) \tag{2.4}$$

The candidate hidden state $\tilde{\mathbf{h}}_t$ uses the reset gate to drop irrelevant past information, while the final hidden state is a linear interpolation controlled by the update gate:

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t \tag{2.5}$$

This interpolation is the key to GRU's efficiency: the network can decide to copy the previous state $\mathbf{h}_{t-1}$ directly (when $\mathbf{z}_t \approx 0$), preserving long-term memory without the complex machinery of the LSTM. Illustrated in **Figure 2.3(b)**, GRUs typically match LSTM performance with fewer parameters, making them less prone to overfitting on datasets with limited samples.



(a) Long Short-Term Memory (b) Gated Recurrent Unit

Figure 2.3: Comparison of Recurrent Units. (a) Long Short-Term Memory (LSTM) with three gates. (b) Gated Recurrent Unit (GRU) with two gates. Source: Chung et al. [8].

2.4 Attention Mechanisms

Attention mechanisms emerged as a response to the "information bottleneck" in fixed-length sequence encodings, first popularized in neural machine translation by Bahdanau et al. [9]. In traditional Recurrent architectures, the entire input sequence must be compressed into a single context vector $\mathbf{v}$. For long sequences, this compression results in loss of detail, as the vector size is fixed regardless of the input length.

Attention solves this by allowing the model to dynamically search for parts of the source sequence that are relevant to the current prediction. Instead of a static vector, Attention computes a *Context Vector* as a weighted combination of all encoder states $\{\mathbf{h}_t\}_{t=1}^{T}$.

Formally, the system learns an alignment model (or scoring function) $a(\mathbf{s}, \mathbf{h}_t)$ that measures how well the inputs around position t and the output at position i match. These scores are normalized using a Softmax function to produce attention weights α_t:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)}, \quad e_t = a(\mathbf{s}, \mathbf{h}_t) \tag{2.6}$$

The context vector $\mathbf{c}$ is then derived as:

$$\mathbf{c} = \sum_{t=1}^{T} \alpha_t \mathbf{h}_t \tag{2.7}$$

In the context of SLR, temporal attention acts as a learned filter. Sign language videos often contain "non-discriminative" frames, such as the hands returning to a resting position or preparatory movements. By assigning near-zero weights ($\alpha_t \approx 0$) to these frames, the model effectively ignores the noise and focuses its computational capacity on the "key frames" where the semantic meaning of the sign is contained. This improves robustness to variable execution speeds and intra-class variability.

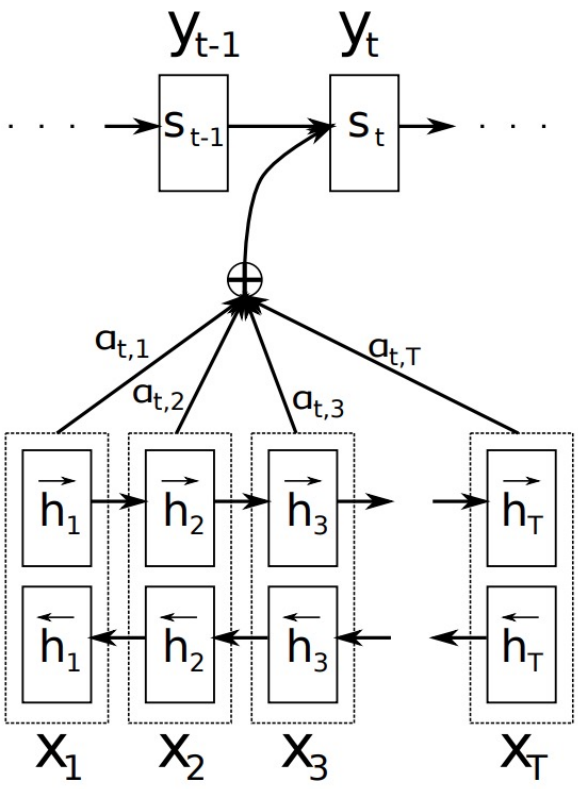


Figure 2.4: Neural Machine Translation architecture with Attention mechanism. The context vector is a weighted sum of encoder annotations. Source: Bahdanau et al. [9].

Chapter 3

Objectives

This chapter defines the purpose of the project, decomposes it into specific and verifiable objectives, and states how each objective is validated in the remainder of the thesis. The objectives describe the research and project outcomes pursued by the work. They are therefore different from the software requirements presented in Section 5.2, which specify the functional and non-functional behaviour expected from the implemented system.

3.1 Main objective

The main objective of this Final Degree Project is to design, implement, and experimentally validate a research prototype for word-level American Sign Language (ASL) recognition from short video clips, using hand-landmark representations and temporal neural models. The purpose is not to build a production-ready translator, but to study whether a compact geometric representation of hand motion can support feasible and reproducible isolated-word sign recognition.

The project also has an accessibility motivation. By exploring automatic recognition of ASL glosses, the work studies how computer vision and deep learning techniques can contribute to assistive communication technologies. The technical system is therefore a means to evaluate a recognition approach, its limitations, and its ethical implications, rather than an end product intended for direct deployment.

3.2 Specific objectives

The main objective is decomposed into six specific objectives. Each one is defined as a measurable project outcome and is later mapped to a validation method in Section 3.3.

- **SO1: Analyse the technical and scientific background of sign language recognition.** This objective covers the relevant foundations of computer vision, pose estimation, MediaPipe Hands, sequence modelling, recurrent neural networks, GRUs, and attention mechanisms, with emphasis on their applicability to word-level sign language recognition.
- **SO2: Define a landmark-based data representation for ASL video clips.** This objective focuses on transforming raw ASL video clips into normalised sequences of three-dimensional

hand landmarks suitable for temporal modelling, while documenting the datasets, preprocessing decisions, and limitations introduced by landmark extraction.

- **SO3: Design and compare candidate temporal models for word-level ASL recognition.** This objective evaluates alternative temporal architectures under a controlled WLASL30 benchmark and identifies the model most suitable for the available data regime.
- **SO4: Validate the selected recognition pipeline through reproducible experimental protocols.** This objective evaluates the selected model using standard classification metrics and protocols that reduce dependence on a single random seed or split, including multi-seed evaluation and stratified cross-validation.
- **SO5: Assess the generalisation and scalability of the recognition pipeline.** This objective tests whether the selected approach transfers to an independent ASL dataset and analyses how performance changes as the WLASL vocabulary size increases.
- **SO6: Contextualise the technical contribution through a runtime proof of concept and ethical analysis.** This objective integrates the recognition pipeline into an end-to-end mobile/server prototype as a proof of concept and analyses the accessibility, privacy, bias, reproducibility, and responsible-deployment implications of the approach.

3.3 Validation methods

Table 3.1 maps each specific objective to the method used to validate it and to the thesis evidence where the committee can verify the result. Detailed numerical results are intentionally left to the Experiments chapter; this section only defines how objective achievement is assessed.

Table 3.1: Validation mapping for the specific objectives.

Objective	Validation method	Evidence in the thesis
SO1	The objective is validated by presenting a structured review of the main techniques needed to understand the proposed system, including convolutional models, pose estimation, MediaPipe Hands, recurrent models, GRUs, and attention mechanisms.	State of the Art chapter (Chapter 2) and the cited literature used to justify the methodological choices.
SO2	The objective is validated by documenting the pipeline that converts ASL videos into normalised landmark sequences and by describing the datasets, extraction process, preprocessing steps, and data-quality constraints.	System design and dataset description in Section 5.1 and Section 5.3; methodological pivot in Section 6.2.
SO3	The objective is validated by comparing candidate temporal architectures on the same WLASL30 benchmark using Top-1, Top-5, Macro-F1, and Balanced Accuracy, and by selecting the strongest model under the controlled comparison.	Architecture selection in Section 6.3 and Table 6.1.
SO4	The objective is validated through the multi-seed protocol, the sequence-length sweep, stratified 5-Fold Cross-Validation, and the reporting of standard metrics across repeated runs.	Experimental methodology in Section 6.1; sequence-length sweep in Table 6.2; WLASL30 validation in Section 6.5 and Table 6.3.
SO5	The objective is validated by retraining and evaluating the same pipeline on MS-ASL50 without architectural changes and by measuring performance degradation across progressively larger WLASL vocabularies.	Cross-dataset validation in Section 6.6, Table 6.4, and Table 6.5; vocabulary scaling in Section 6.7 and Table 6.6.
SO6	The objective is validated by describing the runtime prototype architecture, defining a qualitative validation protocol for live use, and evaluating the ethical and social implications of the system as a research prototype.	Runtime architecture in Section 5.1; qualitative protocol in Section 6.9; ethical and social-impact analysis in Chapter 7.

Chapter 4

Methodology and Work Plan Overview

This chapter describes how the project was actually carried out: which materials and resources were used, which technical methods are applied where, what process governed the day-to-day decisions, in what sequence the work was done, and which constraints shaped every choice. The substance of *what was built* is reserved for the Technical Development chapter (Section 5.1 and following), and the substance of *what was discovered experimentally* is reserved for the Experiments chapter (Section 6.1 and following); this chapter explains the *way of working* that connects them.

4.1 Materials and Resources

The project required four classes of resources: hardware, software stack, datasets, and supervision. Each class is described briefly below and cross-referenced to the chapters where the technical detail lives.

Hardware. All training, evaluation, and landmark extraction were performed on a single local workstation whose specification is the operational environment of Section 5.1.2; the full table of components and version constraints is Table 5.1 and is not repeated here. The relevant element for the work-plan discussion is the GPU: an NVIDIA RTX 3050 Ti Laptop with 4 GB of VRAM. The decision to keep the entire training pipeline on a local GPU rather than on a cloud platform such as Google Colab or AWS was deliberate. Local execution gives bit-for-bit reproducibility between consecutive runs, costs nothing per run (which mattered because the iterative methodology described in Section 4.3 relaunches experiments often), and removes the network-dependent latency that would otherwise penalise short interactive runs. The 4 GB VRAM ceiling is, in turn, the dominant hardware constraint discussed in Section 4.5.

Software stack. The training and evaluation pipeline runs on Python 3.10 with PyTorch 2.x and CUDA 12.x. Pose estimation uses MediaPipe Hands 0.10+, video decoding uses OpenCV 4.8+, and version control is provided by Git with GitHub as remote host. The mobile client is built with

React Native and Expo; the inference server uses FastAPI 0.111 with Uvicorn 0.29. Documentation is authored in LaTeX. The exact version constraints are locked in `pyproject.toml` for the training package and in `04_server/requirements.txt` for the inference server, so that the operational environment is reproducible from a clean clone of the repository (cross-reference: Section 5.1.2 and the reproduction guide at `docs/reproducibility_guide.md`).

Datasets. Six datasets are used across the project: EgoHands and the Kaggle Hand Dataset for hand localisation, Sign Language MNIST and the ASL Alphabet for letter classification, and WLASL and MS-ASL for word recognition. Each dataset is formally defined in Section 5.3, where attribute schema, licence considerations, and the role of each dataset in the experimental pipeline are described. They are listed here only to make the resource inventory complete; the work-plan discussion treats the datasets as external inputs whose acquisition, audit, and preparation costs are accounted for in the corresponding phases of Section 4.4.

Supervision and consultation resources. The project was supervised by Moisés Martínez Muñóz through scheduled tutor meetings held weekly or biweekly depending on the phase of the work. The most consequential technical decisions of the project such as the pivot from pixel-based letter recognition to landmark-based word recognition, the adoption of stratified K-fold cross-validation in place of test-time augmentation, the cross-dataset extension to MS-ASL50 in addition to WLASL, and the introduction of the multi-seed protocol, did not originate as solo decisions but as outcomes of these meetings. The archived meeting notes preserve the directives that triggered each shift, and an internal audit document was used to trace the meeting references cited in this chapter. Consultation resources beyond the tutor consist of the literature reviewed in chapter 2 (notably the WLASL benchmark, MediaPipe Hands, and the Bahdanau attention formulation) and a small number of open-source repositories used as reference implementations: the WLASL benchmark distribution and the MediaPipe Hands toolkit.

4.2 Methods

This section is a compact catalogue of the technical methods the project relies on, presented without re-justification. The theoretical background of each method is the subject of chapter 2, and the empirical evidence for each one is the subject of the Experiments chapter (Section 6.1 and following); the role of this section is to make the inventory explicit so that the rest of the chapter can refer to it.

The perception frontend uses MediaPipe Hands, a top-down two-stage pipeline that first detects the palm and then regresses 21 three-dimensional keypoints per detected hand. The choice to operate on landmarks instead of raw pixels is the central methodological decision of the project and is justified in detail in Section 6.2; the relevant point here is that all downstream methods consume a landmark vector of 63 features per frame.

The temporal model is a bidirectional Gated Recurrent Unit with 256 hidden units per direction and two stacked layers, followed by a Temporal Attention head implemented as an additive (Bahdanau-style) softmax over the per-frame BiGRU outputs. A linear classifier projects the attention-weighted context vector to the gloss vocabulary. This architecture is referred to throughout the document as the GRU+Attention model and totals approximately 2.26 million parameters at the configurations reported in Section 6.1. Two alternative architectures appear in the comparison tables of Section 6.3: a plain bidirectional GRU baseline and a Transformer encoder with four layers and eight heads.

They share the input pipeline and the loss function with the champion model and differ only in the sequence-modelling block.

Optimisation is performed with AdamW at learning rate 1×10^{-3}, weight decay 5×10^{-4}, and label smoothing 0.1 on the cross-entropy loss. The learning-rate schedule is CosineAnnealingLR. Early stopping monitors the validation Top-1 accuracy with patience 30 epochs and a maximum run length of 150 epochs. Skeletal augmentation during training combines rotation, scaling, translation, additive Gaussian noise, and horizontal flip, applied per clip on the landmark coordinates rather than on pixels.

Evaluation uses two complementary protocols. Stratified 5-Fold cross-validation runs on the WLASL30 vocabulary, with the dataset rebuilt by filtering WLASL100 down to the same 30 classes (332 samples) and the folds drawn so that the per-class distribution is preserved. The multi-seed protocol runs the same configuration with three random seeds (42, 123, 2026) and reports the Top-1, Top-5, Macro-F1, and Balanced Accuracy metrics as mean $\pm$ standard deviation. Test-time augmentation was deliberately excluded from the protocol after supervisory review identified it as a source of biased inflation; the rationale is recorded in Section 6.1.

Reproducibility is a method as much as a property. Every training and evaluation run produces an `experiment_manifest.json` that captures the git commit, the resolved configuration, the hardware identifier, the wall-clock time, and the final metrics. Hyperparameters and dataset paths live in YAML files under `configs/`, never in code, with a base configuration defining the defaults shared across experiments and per-experiment files overriding only the fields they need. The full reproduction procedure is documented in `docs/reproducibility_guide.md`.

The exploratory phase of the project relied on two methods that no longer carry results in the Experiments chapter but are listed here for completeness. The first is YOLOv8n hand detection, used during Phase B (see Section 4.4) to localise hands from raw RGB frames. The second is CNN-based pixel classification of static ASL letters. Both methods reached working accuracy on their respective tasks and both were deprecated when the project pivoted to landmark-based word recognition; Section 6.2 explains why.

4.3 Methodology

The project follows an iterative, experiment-driven methodology rather than a fixed waterfall plan, because the technical direction shifted substantially after the early exploratory phase and any plan that did not allow for that shift would have been discarded in February. The methodology is not borrowed from CRISP-DM, Scrum, or any other process-software framework: those frameworks were considered, but the development log documents a different and simpler pattern, applied repeatedly across the lifetime of the project. That pattern is the basis of the methodology described here.

Each meaningful step of the project is organised as a four-stage loop. The loop starts with a *hypothesis* that is verifiable in principle: for example, "landmark-based representation outperforms pixel-based representation on WLASL30 within the available data budget," or "a sequence length of 64 frames produces lower seed-to-seed variance than 32 or 96." The loop continues with *implementation*: the relevant code changes are made in the `src/slr/` package, augmented with whichever new YAML configuration the experiment requires, and committed to git so that the run is anchored to a specific repository state. The loop then runs the *experiment*: the training script is launched, the manifest is written, and the metrics are persisted alongside the model checkpoints in the experiment

artifact directory. Finally, the loop closes with *reflection*: a devlog entry records what was attempted, what the result was, whether the hypothesis was confirmed, and what the next step should be. If the hypothesis fails, the next iteration starts from a new hypothesis informed by the failure rather than from the original plan.

The pivot recorded in Section 6.2 is the most consequential application of this loop. The project began with a pixel-based pipeline that targeted hand detection with YOLOv8n followed by per-frame letter classification. Hand detection reached approximately 98 % mAP@0.5 on the joint EgoHands+Kaggle test split and was, by any reasonable measure, working. Letter classification was equally tractable on the static-frame benchmarks. The combined pipeline, however, surfaced a structural observation that no incremental tuning would have removed: the bottleneck of the system was not localising the hand but representing the dynamic content of the sign. The four-stage loop responded to that observation by retiring the pixel-based recognition stage and replacing it with the landmark-based word-recognition pipeline that is the subject of every subsequent chapter. The pivot was a methodological consequence, not an arbitrary scope change: the iterative loop is what made the pivot possible, and the development log makes the rationale auditable in retrospect.

Two further properties of the methodology are worth making explicit because they show up repeatedly in the Technical Development and Experiments chapters that follow. The first is the role of tutor meetings as control points. The most impactful technical decisions of the project were made jointly with the supervisor at scheduled meetings rather than unilaterally between iterations. These decisions included removing test-time augmentation from the evaluation protocol after supervisory review, restructuring the State of the Art chapter as a hierarchical CNN→RNN→LSTM→GRU theory progression, adopting Google's AI Principles during the ethics review phase, and splitting the experimental work into quantitative and qualitative tracks during the final experimental review. The meetings act as low-frequency external feedback that the four-stage loop, on its own, would not provide.

The second property is the introduction of the multi-seed protocol described in Section 4.2. The protocol is a direct response to a tutor concern about reproducibility raised in early April 2026: a single-seed Top-1 figure on the 34-sample WLASL30 test split is bounded below in resolution by the granularity of one misclassification (≈2.94 pp), and a single number at that resolution is indistinguishable from initialisation noise. The methodology absorbed the concern by adding a new "aggregate over three seeds" step to every headline experiment going forward and by retroactively applying it to the sequence-length sweep and to MS-ASL50. This is a concrete example of the iterative methodology incorporating supervision: the protocol that distinguishes a defensible thesis number from a noisy point was added in response to a single conversation, propagated through the codebase by adding a small number of YAML files, and reflected in Section 6.1 as the statistical backbone of the experimental claims.

Documentation is treated as a continuous side effect of the loop rather than as a separate phase. The development log contains one entry per meaningful session, dated, scoped to a specific session name, and indexed centrally. The log is the primary source from which the work plan in Section 4.4 is reconstructed; without it, the chronology of the project would be unrecoverable from the code alone.

4.4 Work Plan

The executed work plan was reconstructed from the original proposal, recorded development notes, experiment artifacts, and supervision notes. The plan is organised into seven phases (A–G). The first two phases correspond closely to the proposal-era plan, while the remaining phases reflect the methodological pivot from static, pixel-based recognition to landmark-based word-level ASL recognition.

Phase A: Project framing and proposal. The project formally started in the first week of January 2026 with problem definition, preliminary literature review, scope selection, and proposal preparation. This phase defined the accessibility motivation, the initial system architecture, the first work schedule, and the research direction. No model training was carried out at this stage; the main output was the approved project proposal and its initial planning material.

Phase B: Hand detection and static letter recognition. The initial technical work explored the proposal's first recognition route: localising hands in images and classifying static ASL letters. A YOLOv8n detector was trained on a combined EgoHands and Kaggle Hand Dataset corpus, and CNN-based classifiers were explored on static ASL alphabet datasets. These components reached working performance on their own tasks, but they also revealed a limitation in the original direction: the central research challenge was not hand localisation, but representing the temporal structure of complete signs.

Phase C: Landmark pivot and WLASL30 baseline. In late January and February 2026, the project pivoted from pixel-based letter recognition to word-level recognition using MediaPipe hand landmarks. WLASL30 was built as the first controlled benchmark, and several sequence models were compared under the same landmark input representation. The GRU+Attention architecture emerged as the strongest baseline because it combined a recurrent temporal model with an attention mechanism able to focus on the most informative frames. After supervisory review, test-time augmentation was excluded from the final evaluation protocol, and the WLASL30 result was strengthened through stratified K-fold validation.

Phase D: Vocabulary scaling and ethical framework. Once the GRU+Attention model had been selected, the next phase tested how the approach behaved as the WLASL vocabulary increased beyond the initial 30-class setting. The model was retrained on progressively larger WLASL subsets to quantify the effect of vocabulary size and per-class data scarcity. In parallel, the ethical and social-impact chapter was developed, including the accessibility motivation, privacy implications of landmark-based representations, bias considerations, reproducibility, and the responsible-deployment limits of the prototype.

Phase E: Runtime prototype. In April 2026, the technical work expanded from offline training to a runtime proof of concept. A mobile client and FastAPI server were integrated so that short clips could be captured, sent for inference, and returned as Top-K word predictions. Additional prototype features included text-to-speech, model selection from the client, and a heuristic continuous-prediction demo mode based on overlapping windows. This phase demonstrates end-to-end feasibility, but it is not treated as the main research result; the quantitative contribution remains the landmark-based recognition pipeline and its validation.

Phase F: Refactoring and cross-dataset validation. The notebook-based experimental code was consolidated into a modular Python package, with configuration-driven training, structured experiment manifests, reusable evaluation scripts, and preserved checkpoints and plots. This refactor improved reproducibility and made it possible to rerun the same pipeline across datasets. MS-ASL50 was then added as an independent validation benchmark, and the final evaluation protocol incorporated multi-seed runs, a sequence-length sweep, cross-dataset validation, and comparison against the WLASL results.

Phase G: Documentation and qualitative validation design. The final phase focused on thesis writing, result consolidation, figure and table preparation, and the design of a qualitative validation protocol for the runtime prototype. The experimental chapter was organised into quantitative and qualitative tracks: the quantitative track reports completed model evaluations, while the qualitative track defines the participant protocol and analysis plan for future usability evidence. This phase remains open at the time of this intermediate submission and will be completed with the final thesis revision and qualitative data integration.

Figure 4.1 summarises the executed work plan. The chart follows the seven phases described above and highlights the main deviation from the initial proposal: the pivot from static letter recognition to landmark-based word-level recognition.

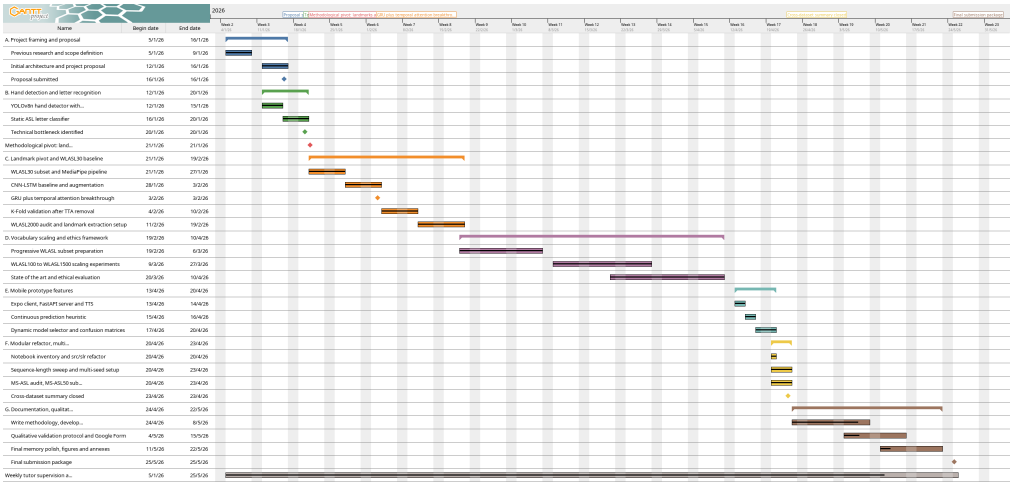


Figure 4.1: Project Gantt chart showing the executed work plan grouped into seven phases, from project framing and initial hand-detection experiments to landmark-based word recognition, cross-dataset validation, prototype integration, and final documentation. Source: self-created with GanttProject.

4.5 Constraints and Limitations

This section catalogues the constraints that shaped the work plan and the methodology rather than the experimental results. Limitations of the experimental conclusions themselves, including standard deviations, test-set granularity, and single-architecture cross-dataset validation, are treated separately in Section 6.8.

The first constraint is time. The project was developed in parallel with a software-engineering internship, which restricted the number of contiguous hours that could be dedicated to it in any given week. The iterative methodology described in Section 4.3 is in part a response to this constraint: a four-stage loop with an explicit devlog entry per session is what makes it possible to resume the project after a multi-day pause without losing the local context. Without the log, the cost of re-entering a problem after an interruption would have dominated the cost of solving it.

The second constraint is hardware. Every training and evaluation run executed on a single NVIDIA RTX 3050 Ti Laptop with 4 GB of VRAM. This ceiling propagated into multiple design choices: `max_frames` = 64 was selected partly because longer sequences exceeded the memory budget at batch size 32, the GRU+Attention model was sized at approximately 2.26 million parameters rather than the parameter counts typical of large Transformers, and the Transformer baseline that does appear in Section 6.3 uses a relatively narrow $d_{\text{model}} = 256$ for the same reason. A cloud GPU was considered and rejected: per-run cost predictability and bit-for-bit reproducibility between consecutive runs were judged more valuable than the additional VRAM.

The third constraint is data availability. Both word-recognition datasets used in the project are reconstructed from external sources rather than distributed directly, and both have lost a measurable fraction of their original clips. WLASL relies on URLs that have decayed since the dataset was published, and the MediaPipe detection rate on the surviving WLASL clips is approximately 69 %, which limits the effective per-clip information content. MS-ASL is reconstructed from per-instance YouTube URLs, and the reconstruction audit records a download success rate of 65.9 % (16,816 of 25,513 clips). The MS-ASL50 subset, rather than the more conventional MS-ASL30, is a direct response to this constraint: only six of the top-30 MS-ASL classes meet the strict 30/5/5 train/val/test threshold from current YouTube state, so an MS-ASL30 reproduction would have been statistically indefensible (cf. Section 6.6).

The fourth constraint is evaluator availability. Until the qualitative validation protocol described in Section 6.9 executes, the author is the only evaluator and the only experimenter on the project. This is a standard configuration for an undergraduate final-degree project and is acceptable for the quantitative work, where the evaluation is fully automated and reproducible from a manifest. It does, however, mean that the cognitive bias of the experimenter is uncontrolled at the design level. The qualitative protocol is structured so that the analysis plan is fixed in advance of data collection, in the spirit of pre-registration, precisely to mitigate this bias once the protocol runs.

The fifth constraint is linguistic scope. The project validates a single sign language, American Sign Language, and uses ASL exclusively for both the WLASL and MS-ASL benchmarks. The pipeline contains no ASL-specific assumptions, and every component that consumes landmarks could in principle be retrained on a different sign language given a comparable benchmark. The validation has not been done, however, and any claim about transfer to Spanish Sign Language, British Sign Language, or any other variety is therefore future work.

The sixth and most general constraint is that the iterative, experiment-driven methodology described in Section 4.3 is, by construction, incompatible with the literal execution of the Gantt that the project proposal recorded in January 2026. The Gantt scheduled hand localisation, static handshape recognition, dynamic sign recognition, temporal fusion and decoding, language generation, mobile prototype, component integration, and validation tests as a sequence of rigid intervals; the methodology actually applied retired the Static Handshape Recognition track in favour of word-level recognition, replaced the proposed PaliGemma language-generation step with a direct gloss-to-text mapping, and folded the Temporal Fusion stage into the GRU+Attention model. This divergence is not a planning

failure but a deliberate property of the methodology. The tutor meeting notes document the moments at which each scope change was approved, and the devlog and meeting notes together act as the audit trail that links the proposal-era plan to the executed plan.

Chapter 5

Technical Development

This chapter describes the engineering side of the project: how the recognition system is structured into independent stages, which functional capabilities each stage provides, in which environment the system runs, and which requirements drove the design. The datasets that feed the pipeline are described in their own section at the end of the chapter.

5.1 System design

The system is structured as a modular three-tier architecture covering data preparation, model training and evaluation, and runtime inference. The three tiers are physically separated (different processes, different machines in the production scenario) but share a single source of truth for data formats and trained model artifacts, so a checkpoint produced in tier two can be served unchanged in tier three. Figure 5.1 shows the overall data flow.

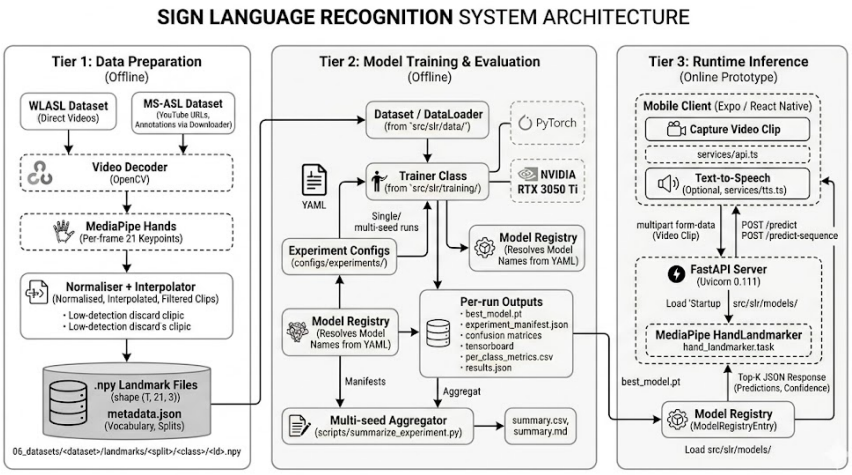


Figure 5.1: System architecture: data preparation, training and evaluation, and runtime inference tiers.

Tier 1: Data preparation pipeline. Raw video material is acquired from the two ASL benchmarks used in the project: WLASL is downloaded directly from the dataset distribution, while MS-ASL is reconstructed by retrieving the original YouTube clips referenced by the dataset annotations and trimming each one with the per-instance `(start_time, end_time)` pair. OpenCV decodes each clip frame by frame, and MediaPipe Hands extracts 21 three-dimensional keypoints per frame. The keypoints are then wrist-centred (the wrist is subtracted from every joint) and length-normalised by the wrist-to-middle-finger distance, so that absolute hand size and frame position no longer carry signal. Frames where MediaPipe fails to detect a hand are linearly interpolated from neighbouring valid frames; clips whose detection rate falls below 0.3 are discarded. Each surviving clip is serialised as a per-clip `.npy` array of shape $(T, 21, 3)$ alongside a `metadata.json` recording the dataset name, gloss vocabulary, split assignments, and the source-video identifiers. This stage is implemented in `src/slr/data/` and exposed through the `scripts/extract_landmarks_*.py` entry points.

Tier 2: Training and evaluation. The Python package `src/slr/` owns the model definitions (`models/` contains `gru_baseline`, `gru_attention`, and `transformer`; a registry resolves the name from a configuration field), the training loop (`training/trainer.py`), the evaluator (`evaluation/`), and the experiment-tracking machinery (`tracking/manifest.py`, `tracking/git_utils.py`). All hyperparameters and dataset paths live in YAML files under `configs/`, never in code; a base configuration (`configs/base/gru_attention.yaml`) declares the defaults shared across experiments, and each experiment YAML overrides only the fields it needs. Every run produces a self-contained directory containing the best checkpoint (by validation Top-1), the training history, confusion matrices (raw and row-normalised), per-class metrics, and an `experiment_manifest.json` that captures git commit, hardware, runtime, hyperparameters, and final metrics. The same trainer supports the multi-seed protocol used to report mean and standard deviation in the experiments chapter.

Tier 3: Runtime inference (prototype). The mobile client lives in `03_app/signtotext-expo/` and is built with React Native and Expo; it captures a short video clip with the device camera and posts it to the inference server through a thin REST client (`services/api.ts`). The server lives in `04_server/main.py`, is implemented with FastAPI, and exposes two endpoints: `POST /predict` for isolated-word inference and `POST /predict-sequence` for a heuristic continuous-sequence variant. On reception, the server runs the same MediaPipe extraction used in tier one, loads the requested model from a curated registry, returns the Top-K predictions with confidence scores, and optionally hands the Top-1 label to a text-to-speech service on the client (`services/tts.ts`). The server reuses the `src/slr/` package directly, so the inference code path is identical to the evaluation code path and no model surgery is needed at deployment time.

5.1.1 Description of functional features

The system exposes six high-level capabilities. Each one is justified by the technical decision it embodies, so this list doubles as a record of why the architecture took its current shape.

- **F1: Landmark-based input pipeline.** Each frame is reduced from a 224×224 RGB tensor to 63 floating-point landmark coordinates, a roughly 2,389$\times$ dimensionality reduction. The choice removes background and skin-tone signal at source, which both shrinks the model and disconnects recognition accuracy from appearance bias.
- **F2: Multi-seed training protocol.** The trainer supports running the same configuration across multiple random seeds and aggregating the metrics with mean and standard deviation. Without this protocol, a single-seed Top-1 figure on a small test set is indistinguishable from noise; with it, a 2–3 percentage-point standard deviation can be reported alongside every headline number.
- **F3: Architecture-agnostic training loop.** The same `Trainer` accepts `gru_baseline`, `gru_attention`, and `transformer` via a registry lookup. This makes the architecture comparison reproducible: only the model field of the YAML changes between runs, every other variable stays fixed.
- **F4: Cross-dataset validation.** The same pipeline runs unmodified on WLASL and on MS-ASL. Only the dataset path and the gloss vocabulary differ between the two configurations; nothing in the model, optimiser, or loss is dataset-specific, which is what makes the cross-dataset comparison fair.
- **F5: Reproducible experiment tracking.** Every run writes an `experiment_manifest.json` that captures git commit, dataset metadata, hyperparameters, hardware, runtime, and final metrics in one place. A future user can reconstruct the exact configuration of any historical run from this single file.
- **F6: Top-K predictions in inference.** The runtime API returns Top-1 and Top-5 candidates with confidence scores instead of forcing a single answer. This matches the way the model itself is evaluated and lets the client surface a shortlist when the Top-1 confidence is low.

5.1.2 Operational Environment

Training and landmark extraction were performed on the local development workstation; the runtime prototype was tested with a paired mobile client and an HTTP server. Table 5.1 lists the components

of the operational environment and the version constraints under which the results in this thesis were produced.

Table 5.1: Operational environment for training, landmark extraction, and runtime inference.

Component	Specification
Operating system	Ubuntu 24.04 LTS (training and server); cross-platform (mobile client).
CPU	Multi-core x86_64 (local laptop).
GPU	NVIDIA RTX 3050 Ti Laptop, 4 GB VRAM, CUDA 12.x.
Python	3.10 (project); 3.11 supported via `requires-python = ">=3.10"`.
Deep learning framework	PyTorch 2.x with CUDA support.
Pose estimation	MediaPipe Hands 0.10+ (`HandLandmarker`).
Video processing	OpenCV 4.8+.
Mobile client framework	React Native + Expo.
Backend	FastAPI 0.111 with Uvicorn 0.29.
Persistence	JSON manifests, `.npy` landmark files, `.pt` PyTorch checkpoints.
Version control	Git.

The development environment is reproducible through the package metadata declared in `pyproject.toml` (editable install with `pip install -e .[landmarks]`) and the dedicated server requirements declared in `04_server/requirements.txt`. The end-to-end reproduction procedure for the headline experiments, including environment setup, dataset preparation, training commands, and evaluation, is documented in `docs/reproducibility_guide.md`.

5.2 Requirements

5.2.1 Description of requirement attributes

For the textual description of the different identified requirements, a set of attributes has been selected to describe each requirement. Below is a description of the meaning of each attribute used:

- **Code**: A unique abbreviated identifier for the requirement. It is constructed using the requirement type code followed by a hyphen and three digits. Requirements are divided into functional and non-functional requirements. The codes used are **FR** for Functional Requirements and **NFR** for Non-Functional Requirements. For example: `FR-001`.
- **Name**: Extended identification of the requirement.
- **Description**: A basic description of the identified requirement.
- **Source**: Indicates the source through which the requirement was identified. This value usually corresponds to one or more use case codes.
- **Necessity**: Determines the degree of implementation required for the requirement. The possible values are:

  - **Essential**: The requirement must be implemented.
  - **Desirable**: It is preferable to implement the requirement, but it is not mandatory.
  - **Optional**: The requirement may be implemented, but it is neither important nor mandatory.
- **Priority**: Defines the importance of the requirement, allowing the order in which requirements are included in the design and implementation process to be established. The possible values are:
  - **High**: The requirement must be implemented in the early stages of development.
  - **Medium**: The requirement should be implemented once high-priority requirements have been completed.
  - **Low**: The requirement should be implemented in the final stages of development. These requirements do not affect the correct functioning of the system.
- **Stability**: Defines the stability of the requirement throughout the software life cycle. This indicates whether the requirement may be modified during the life cycle. The possible values are:
  - **Stable**: The requirement will not change during the system life cycle.
  - **Unstable**: The requirement may change during the system life cycle.
- **Verifiability**: Defines the degree to which a requirement can be verified, that is, the extent to which it is possible to confirm that the requirement has been implemented in the developed system. The possible values are:
  - **High**: It is possible to verify that the requirement has been implemented. These requirements usually correspond to core system functionalities.
  - **Medium**: It is possible to verify that the requirement has been implemented, but it requires complex testing or inspection of the source code.
  - **Low**: It is difficult, or in some cases impossible, to verify whether the requirement has been implemented.

5.2.2 Functional Requirements

The system specifies four functional requirements covering the core recognition pipeline (FR-001, FR-002), the experiment-tracking infrastructure (FR-003), and statistical validation through multi-seed training (FR-004).

System requirement			
ID	FR-001	**Source**	Pipeline design
Name	Landmark extraction from video clips		
Description	The system shall extract 21 three-dimensional hand landmarks per frame from each input video clip using MediaPipe Hands, producing a per-clip array of shape $(T, 21, 3)$ stored as a `.npy` file alongside dataset metadata.		
Necessity	Essential	**Priority**	High
Stability	Stable	**Verifiability**	High

Table 5.2: FR-001: Landmark extraction from video clips

System requirement			
ID	FR-002	**Source**	Project objective
Name	Word-level sign classification		
Description	The system shall classify input video clips into one of the N predefined ASL word classes for the active dataset, returning Top-1 and Top-5 predictions accompanied by confidence scores.		
Necessity	Essential	**Priority**	High
Stability	Stable	**Verifiability**	High

Table 5.3: FR-002: Word-level sign classification

System requirement			
ID	FR-003	**Source**	Reproducibility requirements
Name	Reproducible experiment tracking		
Description	Every training run shall generate a structured manifest capturing dataset version, hyperparameters, git commit, hardware, runtime, and final metrics in a single JSON file, sufficient to reconstruct the run configuration without consulting external sources.		
Necessity	Essential	**Priority**	High
Stability	Stable	**Verifiability**	High

Table 5.4: FR-003: Reproducible experiment tracking

System requirement			
ID	FR-004	**Source**	Statistical validation
Name	Multi-seed training support		
Description	The training subsystem shall support running the same configuration across multiple random seeds and aggregating the resulting metrics with mean and standard deviation, so that every reported result can be accompanied by an uncertainty estimate.		
Necessity	Essential	**Priority**	Medium
Stability	Stable	**Verifiability**	High

Table 5.5: FR-004: Multi-seed training support

5.2.3 Non-functional Requirements

Three non-functional requirements characterise quality attributes of the system: runtime performance (NFR-001), code quality and modularity (NFR-002), and documentation completeness (NFR-003).

System requirement			
ID	NFR-001	**Source**	User experience
Name	Inference latency		
Description	End-to-end inference for a single video clip shall complete in under two seconds on the target deployment hardware (mobile client recording the clip, server running landmark extraction and model prediction).		
Necessity	Desirable	**Priority**	Medium
Stability	Unstable	**Verifiability**	Medium

Table 5.6: NFR-001: Inference latency

System requirement			
ID	NFR-002	**Source**	Maintainability
Name	Modular code architecture		
Description	The codebase shall be organised as a Python package with separate submodules for data, models, training, evaluation, tracking, and utilities, supporting independent unit testing of each submodule.		
Necessity	Essential	**Priority**	High
Stability	Stable	**Verifiability**	High

Table 5.7: NFR-002: Modular code architecture

System requirement			
ID	NFR-003	**Source**	Academic rigor
Name	Documentation completeness		
Description	The repository shall include a reproducibility guide that enables a third party to replicate the headline experimental results with documented commands and dependency declarations.		
Necessity	Essential	**Priority**	Medium
Stability	Stable	**Verifiability**	Medium

Table 5.8: NFR-003: Documentation completeness

5.3 Datasets

This project spans two recognition tasks with fundamentally different data requirements: hand localization from still images, and word-level sign language recognition from video. As a consequence, six distinct datasets are employed across the full experimental pipeline. This section formalizes the properties of each dataset by first defining a common set of descriptive attributes, then presenting a unified comparative table, and finally providing individual descriptions of each dataset in context.

5.3.1 Dataset Attribute Definitions

To ensure a consistent and unambiguous characterization of every dataset, the following attributes are used in Table 5.9. Each attribute captures a distinct dimension relevant to dataset selection, reproducibility, and the interpretation of experimental results:

- **ID**: A unique alphanumeric identifier assigned to each dataset within this thesis. The prefix `DS-` followed by a two-digit index allows unambiguous cross-referencing in subsequent chapters when reporting experimental configurations and results.
- **Name**: The official or widely-used name of the dataset as it appears in the literature or on its hosting platform.
- **Year**: The year of original publication or first public release. This contextualizes the dataset within the chronological development of the field and helps evaluate whether it reflects the current state of the art in data collection methodology.
- **Task**: The specific sub-task for which the dataset was designed. Three task categories appear in this project:
  - **Hand Detection**: the dataset provides annotated bounding boxes or segmentation masks marking the location of hands in images.
  - **Letter Recognition**: the dataset provides labeled static images of individual American Sign Language (ASL) letter signs.
  - **Word Recognition**: the dataset provides labeled video clips of complete ASL word signs, requiring temporal modeling.
- **Samples**: The total number of annotated data instances. For image datasets, this equals the number of images; for video datasets, it equals the number of individual video clips. Where a subset of the full dataset was used in this project, this is noted in the individual descriptions.
- **Classes**: The number of distinct output categories. For hand detection datasets this is always 1 (the single *hand* class); for sign language datasets it corresponds to vocabulary size, meaning the number of distinct letters or word glosses.
- **Format**: The storage format and annotation type of the raw data. This attribute specifies whether instances are static images or video clips, the image resolution where relevant, and the form of ground-truth annotation (e.g., bounding box coordinates in XML, pixel-level segmentation masks, image class labels, or video-level class labels).
- **Input Representation**: Whether the models trained on this dataset consume *pixel-based* data (raw RGB or grayscale image/video frames) or *landmark-based* data (structured 3D keypoint coordinates extracted by a pose estimator such as MediaPipe). Pixel-based inputs preserve full visual detail but are high-dimensional and sensitive to background, lighting, and signer appearance. Landmark-based inputs are compact, viewpoint-invariant, and more robust to visual variability, at the cost of depending on a reliable keypoint extractor.

5.3.2 Comparative Overview

Table 5.9 provides a high-level comparison of the six datasets used in this project. Datasets DS-01 and DS-02 support the hand localization pipeline described in later experimental sections. Datasets DS-03 and DS-04 were used in the initial letter-level classification phase of the project, which was

subsequently superseded by the word recognition approach. Datasets DS-05 (WLASL) and DS-06 (MS-ASL) are the primary benchmarks for word-level recognition. WLASL is the dataset on which the architecture was tuned; MS-ASL was added in a later phase to validate that the same pipeline transfers to an independent ASL corpus with a different signer distribution.

Table 5.9: Comparative overview of all datasets used in this project.

ID	Name	Year	Task	Samples	Classes	Format
DS-01	EgoHands	2015	Hand Detection	4,800	1	Images + Seg. Masks
DS-02	Kaggle Hand Dataset	2023	Hand Detection	∼4,200	1	Images + BBox (XML)
DS-03	Sign Language MNIST	2017	Letter Recog.	34,627	24	28×28 Grayscale CSV
DS-04	ASL Alphabet	2018	Letter Recog.	87,000	29	200×200 RGB JPG
DS-05	WLASL	2020	Word Recog.	21,083	2,000	Video Clips (RGB)
DS-06	MS-ASL	2018	Word Recog.	25,513	1,000	Video Clips (RGB, YouTube)

5.3.3 Hand Detection Datasets

Datasets DS-01 and DS-02 are used jointly to train the hand localization module based on YOLOv8. A central design motivation for combining two datasets is improved generalization: a detector trained exclusively on egocentric footage tends to fail when the camera is positioned at eye level or beyond arm's reach, while a detector trained only on third-person images may struggle with first-person perspectives. By merging the two sources, the training corpus covers a significantly wider range of viewpoints, backgrounds, and hand appearances. After merging, duplicate removal, and a stratified train/validation/test split, the combined training set comprises approximately 5,223 annotated images.

DS-01: EgoHands [10]

EgoHands is an egocentric hand detection dataset published by Indiana University in 2015 as part of research on activity recognition in close-proximity social interactions. It consists of approximately 4,800 labeled frames extracted from 48 video recordings captured from a GoPro camera mounted on the chest of an observer watching another person. Ground-truth annotations are stored as pixel-level segmentation polygons in MATLAB `.mat` files; during preprocessing, these polygons were converted to axis-aligned bounding boxes in YOLO format. The dataset covers diverse interaction scenarios, including card games, chess, and jigsaw puzzles, with two hands simultaneously visible in most frames, providing challenging overlap and occlusion cases. Because all images are taken from a first-person perspective, the visual distribution of EgoHands is inherently biased toward the observer's own hand. This limitation motivates the inclusion of DS-02 to extend viewpoint diversity. The input representation is pixel-based: models consume raw RGB crops.

DS-02: Kaggle Hand Dataset [11]

The Kaggle Hand Dataset is a collection of annotated hand images published on the Kaggle platform, providing multi-viewpoint coverage that is complementary to the egocentric nature of DS-01. Images depict hands in third-person perspectives with diverse backgrounds, lighting conditions, skin tones, and partial occlusions. Ground-truth bounding boxes are provided in Pascal VOC XML format; during preprocessing, coordinates were normalized to the YOLO format used by the detection pipeline. The dataset contains approximately 4,200 labeled images. The joint training corpus formed by DS-01 and DS-02 yielded a YOLOv8n model achieving approximately 98% mAP50 and 87% Mean IoU on the held-out test split, demonstrating the effectiveness of the multi-source approach. The input representation is pixel-based.

5.3.4 Letter Recognition Datasets

Datasets DS-03 and DS-04 were used during the initial phase of the project, when the recognition target was the static letter-level ASL alphabet. Both datasets cover 24 of the 26 ASL letters; the letters *J* and *Z* are excluded in both cases because they require hand motion and cannot be represented by a single static image. The two sources were merged and unified into a single training corpus for letter classification experiments. Although this phase was later superseded by the more ambitious word-level recognition approach, these datasets remain relevant because they established the viability of data-driven letter classification and informed the design of the preprocessing pipeline. The input representation for both datasets is pixel-based.

DS-03: Sign Language MNIST [12]

Sign Language MNIST mirrors the format of the canonical MNIST handwritten digit benchmark. It contains 27,455 training images and 7,172 test images, for a total of 34,627 samples, each representing one of 24 static ASL letters as a 28×28 grayscale image. Data is distributed in CSV format, with one row per image where each column corresponds to a pixel intensity value in the range $[0, 255]$, plus a label column. The low resolution and grayscale format make this dataset extremely lightweight and straightforward to preprocess, but inherently limit the discriminative information available to the model, as fine-grained finger articulation details are largely lost. The dataset was originally constructed from ASL alphabet photographs rescaled and converted to the MNIST-style format, and is hosted on Kaggle.

DS-04: ASL Alphabet [13]

The ASL Alphabet dataset provides a substantially richer visual input than DS-03. It contains approximately 87,000 images organized into 29 classes: the 24 static ASL letters, the letter *space*, and two additional non-letter classes (*del*, *nothing*). Each class is represented by exactly 3,000 training images captured in a controlled environment at a resolution of 200×200 pixels in full RGB color. The higher resolution and color information enable more discriminative feature learning, particularly for hand texture and illumination cues. For integration with DS-03 into a unified corpus, images were converted to grayscale and resized to a common format, yielding a combined training set of approximately 79,000 images and 7,000 test images covering the 24 shared letter classes.

5.3.5 Word Recognition Dataset

DS-05: WLASL [2]

The Word-Level American Sign Language (WLASL) dataset, published by Li et al. in 2020, is the primary data resource for all word-level recognition experiments in this thesis and one of the largest publicly available ASL word recognition benchmarks to date. WLASL contains 21,083 video instances covering 2,000 ASL glosses (words). Videos were collected from multiple online platforms, including YouTube channels maintained by ASL educators, online courses, and signing dictionaries, and manually verified by qualified ASL experts to ensure correctness of sign production. Each clip shows a single signer performing one complete sign, typically against a plain or low-clutter background, with durations ranging from approximately one to four seconds.

The dataset is partitioned into four progressively larger evaluation subsets: WLASL100 (100 glosses), WLASL300 (300 glosses), WLASL1000 (1,000 glosses), and WLASL2000 (the full 2,000 glosses). This structure enables systematic benchmarking of how model performance scales with vocabulary size. A practical challenge is that WLASL relies on external video URLs; a fraction of clips have become unavailable over time due to deletion or access restriction, so only successfully downloaded videos can be used.

In this project, raw RGB frames are not used as model input. Instead, MediaPipe Hands [1] is applied to each frame to extract 21 three-dimensional hand keypoints, yielding a compact landmark vector of 63 scalar features per frame (21 joints $\times$ 3 coordinates). This representation reduces input dimensionality by a factor of approximately 2,389 compared to raw 224×224 RGB frames, removes background and appearance noise, and provides geometric invariance to signer appearance. The effective detection rate of MediaPipe on WLASL clips is approximately 69%; frames where no hand is detected are linearly interpolated from adjacent detected frames to maintain sequence continuity. Each video is resampled to a fixed sequence length of 64 frames before being fed to the model.

For the experiments described in this thesis, the following WLASL subsets were extracted and used:

- **WLASL30**: The 30 most frequent glosses, used for initial architecture exploration and model selection. Split: 242 training / 56 validation / 34 test videos.
- **WLASL100**: The 100 most frequent glosses, used for K-Fold cross-validation and robust evaluation. Split: 741 training / 162 validation / 99 test videos.
- **WLASL300, WLASL1000, WLASL2000**: Larger subsets targeted for progressive scaling experiments in later stages of the project, where the goal is to assess model accuracy and generalization at increasing vocabulary sizes.

DS-06: MS-ASL [3]

The Microsoft American Sign Language (MS-ASL) dataset, published by Vaezi Joze and Koller in 2018, is a second large-scale ASL benchmark used in this project to validate that the recognition pipeline tuned on WLASL transfers to an independent corpus with a different signer distribution. The original release defines 25,513 video instances over 1,000 glosses, split into 16,054 train / 5,287 validation / 4,172 test samples. Clips are not distributed directly: the dataset ships only annotation JSONs and relies on per-instance YouTube URLs plus `(start_time, end_time)` timestamps and a normalized signer bounding box. The vocabulary is organized into four cumulative subsets (MS-ASL100/200/500/1000) obtained by filtering on the integer label.

Acquiring the raw clips is therefore a non-trivial preprocessing step. Of the 25,513 declared instances, **16,816 were successfully downloaded (65.9% coverage)**, with split-level coverage of 74.0% / 47.5% / 58.0% (train / val / test). The shortfall is caused by YouTube videos being removed or made private over time and is the dominant constraint on which subsets can be reproduced today. A disk-level audit then graded every downloaded clip by integrity and by training viability. Only **6** of the top-30 classes met the strict MS-ASL30 threshold of 30 train / 5 val / 5 test surviving samples; **17** met the MS-ASL50 threshold of 20 / 5 / 5; and **57** met the MS-ASL100 threshold of 20 / 3 / 3. The strict MS-ASL30 cut would discard 24 of the 30 nominal classes, so an MS-ASL30 reproduction was not viable. The chosen alternative is **MS-ASL50**, defined as the 50 classes with the largest train counts that satisfy the 20 / 3 / 3 threshold; this preserves the size-of-vocabulary objective while keeping every class statistically defensible.

The MS-ASL50 subset was processed with the same MediaPipe pipeline used for WLASL: each frame is reduced to 21 three-dimensional hand keypoints (63 features) and stored as a per-clip `.npy` array. After extraction, the final per-split counts are **1,410 train / 306 val / 236 test (1,952 total)**, with 31 source clips dropped because their mean detection rate fell below the 0.3 floor. The mean detection rate across the surviving clips is **83.9%**, substantially above the WLASL figure, consistent with MS-ASL clips being longer and shot under more controlled conditions. Apart from the change of dataset, the GRU + Temporal Attention model consumes the splits without any architectural modification.

Chapter 6

Experiments

This chapter presents the experimental work in the order in which it actually unfolded during the project, from the early hand-detection and letter-classification probes that motivated the methodological pivot, through the architecture comparison and hyperparameter sweep that produced the champion configuration, to the multi-seed and cross-dataset validation that closed the loop. Every result reported below is backed by stored experiment artifacts, including configurations, metrics, plots, trained checkpoints, and an experiment manifest for each run. The detailed repository paths needed to reproduce or audit the results are centralised in the reproducibility guide and internal audit material, so the chapter cites experiment outputs conceptually rather than repeating long file paths in every caption.

6.1 Experimental Methodology

The experimental design rests on two complementary protocols. The first is a **multi-seed protocol** that runs every headline configuration with three random seeds (42, 123, 2026) and reports metrics as mean $\pm$ standard deviation. The motivation is statistical: the WLASL30 test split contains only 34 samples, so a single misclassification moves Top-1 by approximately 2.94 percentage points. A single-seed Top-1 figure on this test set is therefore indistinguishable from initialisation noise, and any claim about the system's behaviour requires aggregating across seeds. The second protocol is **Stratified 5-Fold Cross-Validation** on WLASL100 filtered to the same 30 classes (332 total samples, 5 non-overlapping folds with class distribution preserved); it complements the multi-seed protocol by varying the test split itself rather than only the model initialisation.

Four metrics are reported throughout the chapter. **Top-1 accuracy** is the proportion of test samples for which the highest-scoring prediction matches the ground truth and is the headline figure most often quoted. **Top-5 accuracy** is the proportion of test samples for which the ground truth appears anywhere in the five highest-scoring predictions; it matters in this domain because a real-time captioning system can present a shortlist when the Top-1 confidence is low. **Macro-F1** is the unweighted mean of per-class F1, which exposes degradation on rare classes that Top-1 averages would hide. **Balanced accuracy** averages per-class recall, providing a complementary view of class-imbalance effects.

All training runs share the same hardware and optimisation schedule. Training and evaluation ex-

ecute on a local workstation with an NVIDIA RTX 3050 Ti Laptop GPU (4 GB VRAM, CUDA 12.x). The optimiser is AdamW with learning rate 1×10^{-3}, weight decay 5×10^{-4}, and label smoothing of 0.1 on the cross-entropy loss. The learning rate follows a CosineAnnealingLR schedule. Early stopping monitors validation Top-1 with patience 30 epochs, and the maximum run length is 150 epochs. These values are fixed across every experiment in this chapter unless explicitly noted otherwise.

Test-Time Augmentation (TTA) was deliberately excluded from the evaluation protocol. Earlier exploratory work compared TTA-enabled and TTA-disabled runs on WLASL30 and found that TTA inflates the reported Top-1 without producing a corresponding gain in true generalisation. The decision is methodological rather than incidental: every number in this chapter reflects a single forward pass per test sample.

Part I: Quantitative Evaluation

6.2 Phase 1: Exploratory Work and Methodological Pivot

The first months of the project explored two avenues that were eventually superseded by the landmark + word approach. They are reported here because the lessons drawn from them are the technical justification of the pivot, not because they appear as headline numbers in the final system.

Hand detection with YOLOv8n. An initial sub-system trained YOLOv8n [14] on the joint EgoHands and Kaggle Hand Dataset corpus described in Section 5.3.3 (approximately 5,223 annotated images after stratified train/val/test split and duplicate removal). The model reached approximately **98 % mAP@0.5** and a **Mean IoU of approximately 87 %** on the held-out test split, as verified by the detector evaluation outputs. YOLOv8n was selected after a wider comparison against SSDLite-MobileNetV3 and EfficientDet-D0; the YOLO family won on inference latency at comparable detection accuracy on this dataset.

Letter-level pixel classification. A parallel track trained CNN classifiers on the union of Sign Language MNIST (DS-03) and ASL Alphabet (DS-04). Coverage was limited to 24 of the 26 ASL letters because *J* and *Z* require hand motion and cannot be represented by a single static image. Accuracy on this static-frame classification task was high, but the task was also far simpler than the eventual project goal: predicting one of $\sim$24 hand poses per frame is a different problem from predicting a word from a multi-frame sign.

What the exploration revealed. The bounding-box detector was not the bottleneck, since it worked reliably. The bottleneck was *representation*. Pixel-based recognition has three properties that are problematic at the WLASL scale: (i) it carries appearance signal (skin tone, lighting, background) that is irrelevant to the linguistic content of a sign and that the model has to learn to discount with limited data; (ii) the parameter cost of an image-conditioned sequence model grows quickly, which makes overfitting on a few hundred clips per class very likely; and (iii) the geometry of hand articulation, which is the actual carrier of meaning, is not made explicit to the model.

The pivot to landmark-based word recognition replaced the visual problem with a geometric one: each frame is reduced from a 224×224 RGB tensor to 21 three-dimensional keypoints (63 features), a reduction of approximately $2{,}389\times$ in input dimensionality. The representation discards

background and appearance at the source and exposes the joint geometry directly to a temporal model. The remainder of this chapter studies the resulting pipeline.

6.3 Architecture Selection on WLASL30

WLASL30 (the 30 most frequent WLASL classes; 242 train / 56 validation / 34 test clips) was used as a controlled bench to compare candidate architectures under matched conditions. All five rows in Table 6.1 share the same dataset, the same input pipeline, the same hardware, and the same hyperparameters from Section 6.1, with the exception of the Transformer row whose configuration uses $d_{\text{model}} = 256$, $\text{nhead} = 8$, four transformer layers, $\text{lr} = 5 \times 10^{-4}$, $\text{wd} = 1 \times 10^{-4}$, and dropout 0.3, as recorded in the corresponding experiment manifest.

Table 6.1: Single-split test results of the five architectures evaluated on WLASL30 (34 test samples). Source: self-created from experiment outputs. Standard deviations are not available at this stage; the multi-seed validation of the champion is in Section 6.5.

Architecture	**Top-1 (%)**	**Top-5 (%)**	**Notes**
LSTM (landmarks)	41.18	67.65	Plain LSTM baseline.
GRU (landmarks)	38.24	76.47	Plain bidirectional GRU.
GRU + Enriched features	47.06	79.41	Bone vectors and joint-angle augmentations.
GRU + Attention	**58.82**	**91.18**	Champion; 2.26 M parameters.
Transformer	50.00	73.53	Four layers, 8 heads, $d_{\text{model}} = 256$.

The ordering in Table 6.1 is consistent with the data regime. Plain GRU and plain LSTM are both starved for training data at 242 examples spread over 30 classes; with a comparable parameter budget, the GRU's lower count reduces overfitting at the cost of some ceiling. Adding hand-engineered enriched features (bone vectors, inter-joint angles) gives the model a 9-point absolute Top-1 lift, but the gain disappears once the model is given a Temporal Attention layer that can learn the discriminative time slices on its own. The Attention head dominates because sign language clips contain a small number of *key frames* where the linguistic content is concentrated, surrounded by neutral preparation and recovery frames; weighting the GRU outputs by a learned attention distribution focuses capacity where it matters and tolerates variation in execution speed across signers.

The Transformer row is the most pedagogically interesting. With a parameter budget that is competitive with the GRU+Attention model, the Transformer underperforms on Top-1 (50.00 vs. 58.82) and underperforms more sharply on Top-5 (73.53 vs. 91.18). The result is consistent with the observation in the literature that recurrent inductive biases are an asset in low-data regimes; the Transformer's lack of a built-in sequential prior would plausibly be amortised at WLASL2000 scale, but the experiment in this section does not provide that scale.

The champion configuration, a bidirectional GRU with hidden size 256, two layers, dropout 0.4, followed by a Temporal Attention head, is therefore selected as the default for every downstream experiment in the chapter. Its confusion matrix on the WLASL30 test split is shown in Figure 6.1.

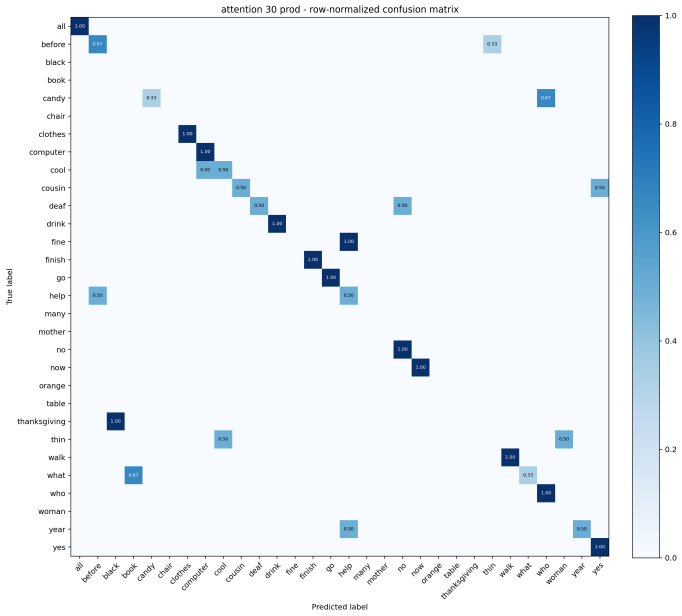


Figure 6.1: Row-normalised confusion matrix of the GRU+Attention champion on the WLASL30 test split (34 samples). Source: self-created from experiment outputs.

The matrix exposes two structural failure modes. The first is class imbalance at evaluation time: several test classes have only one or two samples, so a single error is enough to drop their per-class recall to zero. The second is genuine visual similarity between specific signs: according to the stored experiment outputs, the most common confusions on this split are *candy* $\rightarrow$ *who* and *what* $\rightarrow$ *book*, both at 2 of 3 affected samples, plus single-sample failures on *fine* $\rightarrow$ *help*, *thanksgiving* $\rightarrow$ *black*, and *cool* $\rightarrow$ *computer*.

6.4 Sequence Length Sweep

The maximum sequence length `max_frames` is the only hyperparameter that the model sees as a structural input shape rather than as a regulariser, so its value has to be defended explicitly. Subsampling too aggressively (32 frames) discards temporal dynamics; oversampling (96 frames) introduces padding and noise on shorter clips. A multi-seed sweep distinguishes signal from initialisation variance.

The sweep covers nine runs: 3 seeds (42, 123, 2026) $\times$ 3 sequence lengths (32, 64, 96), with every other variable held fixed at the values in Section 6.1. Aggregated results from the sweep are reported in Table 6.2.

Table 6.2: Sequence-length sweep on WLASL30 with 3 seeds per row. Mean ± standard deviation across seeds. Top-1 standard deviations are bounded below by the test-set granularity (≈2.94 pp per misclassification on 34 samples). Source: self-created from experiment outputs.

Frames	Top-1	Top-5	Macro-F1	Balanced Acc.
32	56.86 ± 6.12	85.29 ± 2.95	50.52 ± 8.27	64.65 ± 7.48
64	58.82 ± 2.94	78.43 ± 4.49	54.88 ± 0.93	67.42 ± 0.76
96	54.90 ± 4.49	81.37 ± 4.49	49.63 ± 6.33	61.36 ± 6.56

The 64-frame configuration produces the most stable training across the three seeds. Macro-F1 standard deviation drops to 0.93 pp, and Balanced Accuracy standard deviation to 0.76 pp, both an order of magnitude tighter than at 32 or 96 frames. The sweep does not select the configuration that maximises any single mean metric, since 32 frames actually edges out 64 on Top-5 (85.29 vs. 78.43), but it does select the configuration whose mean is least sensitive to the seed. The selection criterion is reproducibility, not peak Top-1, and that is the criterion this chapter applies to every downstream choice.

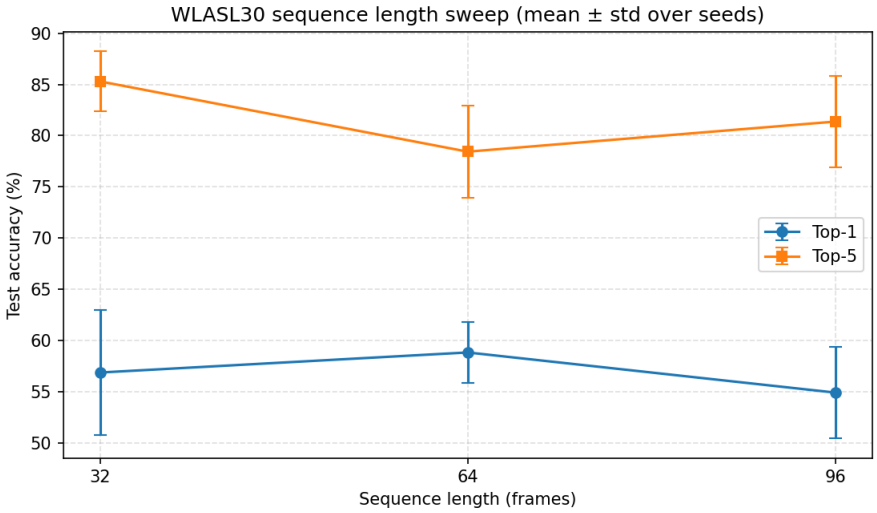


Figure 6.2: Per-metric distribution across the sequence-length sweep (3 seeds at 32, 64, and 96 frames). Source: self-created from experiment outputs.

The 64-frame value is therefore fixed as the default for every downstream experiment in this chapter, including the cross-dataset MS-ASL50 evaluation in Section 6.6. A dedicated sweep on MS-ASL50, where median clip length is approximately 80 frames, would be a natural extension and is acknowledged in Section 6.8.

6.5 WLASL30 Validation

Two independent validation protocols characterise the champion model on WLASL30. The first is the **Stratified 5-Fold Cross-Validation** run on WLASL100 filtered to the same 30 classes (332 total samples), which yields a Top-1 mean of **63.85 %** $\pm$ **4.50 %** and a Top-5 mean of **86.46 %** $\pm$ **2.04 %**. This is the headline thesis number because it averages across both initialisation and split, and because the standard deviations are computed across non-overlapping test partitions. The second is the **multi-seed evaluation** on the fixed WLASL30 split at $max_frames = 64$ extracted from the sequence-length sweep (Table 6.3).

Table 6.3: Per-seed test results of the GRU+Attention champion on the fixed WLASL30 split (34 test samples). Source: self-created from experiment outputs.

Seed	Top-1	Top-5	Macro-F1	Balanced Acc.	Best Epoch
42	55.88	73.53	55.94	68.18	40
123	61.76	79.41	54.20	66.67	29
2026	58.82	82.35	54.49	67.42	27
Mean $\pm$ Std	58.82 ± 2.94	78.43 ± 4.49	54.88 ± 0.93	67.42 ± 0.76	N/A

The two protocols disagree by approximately five percentage points on Top-1 (63.85 vs. 58.82). The disagreement is informative rather than contradictory: 5-Fold CV varies the test partition itself, so the Top-1 average is taken over five non-overlapping test partitions of approximately 66 samples each ($332/5 = 66.4$) and effectively smooths the worst single split; the multi-seed protocol fixes the 34-sample test split and varies only the model initialisation and batch order. Both quantities are real; they answer different questions. The headline number reported elsewhere in the thesis is the 5-Fold mean because it is the more statistically robust of the two and is directly comparable to the K-Fold standard in the SLR literature.

The per-class behaviour of the champion on the fixed split is concentrated on a small number of low-support classes. According to the per-class metric outputs, the worst recall values among supported classes are *fine* (1 sample, recall 0.00), *thanksgiving* (1, 0.00), *thin* (2, 0.00), *candy* (3, 0.33), *what* (3, 0.33), and *cool* (2, 0.50). With one to three test samples per class, even a single misprediction yields a 33–100 pp recall drop. This is a property of the test split, not of the model: the per-class metric can be recovered only by enlarging the evaluation set, which is exactly what the cross-dataset experiment in Section 6.6 achieves.

6.6 Cross-Dataset Validation: MS-ASL50

The pipeline was tuned on WLASL30. To check whether its behaviour is a property of the method or of the benchmark, the same code path was retrained on MS-ASL50 with no architectural changes and no dataset-specific hyperparameter tuning. The only fields that differ between the two configurations are the dataset path and the gloss vocabulary.

6.6.1 Building the MS-ASL50 subset

MS-ASL ships only annotation JSONs and YouTube URLs; the 25,513 declared instances must be retrieved per-clip, and not all of them are still online. The dataset reconstruction audit records that **16,816 of 25,513 clips were successfully downloaded (65.9 %)**, with split-level coverage of 74.0 % / 47.5 % / 58.0 % (train / val / test). The drop-out is dominated by YouTube videos that have been removed or made private since the dataset was published.

The subset selection itself is constrained by data deficiency. The disk-level audit reports that only **6** of the top-30 classes meet the strict 30 train / 5 val / 5 test threshold; **17** meet the 20 / 5 / 5 threshold; and **57** meet the relaxed 20 / 3 / 3 threshold. An MS-ASL30 reproduction is therefore not viable from the locally available data. The chosen alternative is **MS-ASL50**, defined as the 50 classes with the largest train counts among those satisfying the 20 / 3 / 3 threshold. After landmark extraction, 31 source clips are dropped because their mean MediaPipe detection rate falls below the 0.3 floor, leaving **1,410 train / 306 val / 236 test (1,952 total)**. The mean detection rate across surviving clips is **83.9 %**, substantially above the WLASL figure (approximately 69 %), consistent with MS-ASL clips being longer and shot under more controlled conditions.

6.6.2 Cross-dataset results

Both datasets were trained with three seeds (42, 123, 2026). Aggregated metrics are in Table 6.4, with the MS-ASL50 per-seed breakdown in Table 6.5.

Table 6.4: Side-by-side comparison of the GRU+Attention pipeline on WLASL30 and MS-ASL50. Both rows are aggregated over 3 seeds. Source: self-created from experiment outputs.

Dataset	Classes	Train / Val / Test	Top-1	Top-5	Macro-F1
WLASL30	30	242 / 56 / 34	58.82 ± 2.94	78.43 ± 4.49	54.88 ± 0.93
MS-ASL50	50	1,410 / 306 / 236	70.20 ± 2.89	90.54 ± 0.89	68.08 ± 3.44

Table 6.5: Per-seed test results on MS-ASL50 (236 test samples). Source: self-created from experiment outputs.

Seed	Top-1	Top-5	Macro-F1	Balanced Acc.	Best Epoch
42	66.95	89.83	64.22	66.34	55
123	72.46	90.25	70.81	72.75	120
2026	71.19	91.53	69.21	71.58	92
Mean $\pm$ Std	70.20 ± 2.89	90.54 ± 0.89	68.08 ± 3.44	70.22 ± 3.41	N/A

The central observation is that the Top-1 standard deviations on the two datasets are nearly identical (2.94 pp on WLASL30 versus 2.89 pp on MS-ASL50). With $7\times$ fewer test samples on WLASL30, this is the strongest statement the experiment supports: *training stability is a property of the pipeline, not of a specific benchmark*. The pipeline does not need dataset-specific tuning to produce reproducible results on a second ASL corpus.

The absolute Top-1 figure on MS-ASL50 (70.20) is higher than on WLASL30 (58.82), but this is not evidence that MS-ASL is intrinsically easier. Three structural factors account for the gap. First, MS-ASL50 has approximately 6× more training samples (1,410 vs. 242), which is the dominant variable in this data regime. Second, the test set is 7× larger (236 vs. 34), so each individual misclassification moves Top-1 by approximately 0.42 pp instead of 2.94 pp; a finer measurement grid yields more representative values, which usually means closer to the true asymptotic performance. Third, MS-ASL clips are sourced from longer, better-lit YouTube educational material, which is reflected in the higher MediaPipe detection rate. The experiment therefore supports the claim that the pipeline reproduces; it does not support a claim about intrinsic dataset difficulty.

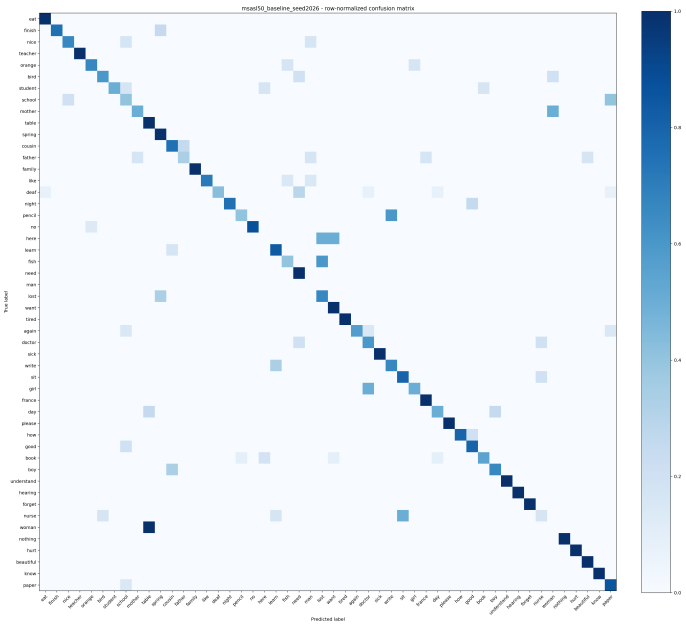


Figure 6.3: Row-normalised confusion matrix on the MS-ASL50 test split for seed 2026 (236 samples). The matrices for the other seeds show the same qualitative structure. Source: self-created from experiment outputs.

The confusion structure, as verified by the stored experiment outputs, is dominated by visually adjacent signs. The largest absolute confusion is *deaf* → *need* (4 of 14 samples, 28.6 %), followed by *pencil* → *write* and *fish* → *lost* (each 3 of 5, 60.0 %), *nurse* → *sit* (3 of 6, 50.0 %), and *mother* → *woman* (2 of 4, 50.0 %). These are recognisable as one-handed, location-similar pairs whose discriminative cue is fine-grained finger shape rather than gross trajectory.

The per-class metrics show three classes with 0.00 recall (*here*, *man*, *woman*, all with three test samples), and six further classes below 50 % recall (*need* at 0.25, *school* at 0.29, *nurse* at 0.33, *lost* at 0.33, *write* at 0.40, *doctor* at 0.43). At the other end, nineteen classes reach perfect recall on this split, including six with five or more support samples (*teacher* 9, *no* 7, *tired* 7, *deaf* 6, *please* 5, *like* 5). The headline Macro-F1 of 68.08 is therefore an average over a strongly bimodal per-class

distribution, which is consistent with the higher Macro-F1 standard deviation (3.44) compared to Top-1 (2.89): per-class recall is a finer-grained signal and amplifies seed-to-seed variation in which classes happen to be learned best.

6.7 Vocabulary Scaling

WLASL exposes progressively larger subsets of its 2,000-class vocabulary. Retraining the champion configuration on each subset quantifies how the system degrades as the recognition problem grows. The experiments in this section all use a single seed (42); multi-seed validation was not performed beyond 30 classes.

Table 6.6: Vocabulary scaling of the GRU+Attention champion on WLASL subsets. The 30-class row is the multi-seed mean from Table 6.4; all other rows are single-seed runs and standard deviations cannot be reported. *Train time* is the total wall-clock time of the training run on the NVIDIA RTX 3050 Ti Laptop GPU described in Section 6.1. Source: self-created from experiment manifests.

Vocabulary	Train / Val / Test	Top-1	Top-5	Best Epoch	Train time (s)
30	242 / 56 / 34	58.82 ± 2.94	78.43 ± 4.49	27–40	N/A
100	795 / 186 / 132	44.70	70.45	99	83
300	1,974 / 474 / 360	28.61	59.72	112	224
500	2,967 / 723 / 561	23.71	49.55	111	311
700	3,879 / 974 / 736	19.57	42.66	57	248
900	4,707 / 1,202 / 880	12.50	33.75	220	881
1,000	5,122 / 1,329 / 938	11.19	31.24	113	602
1,200	5,840 / 1,536 / 1,051	10.56	27.69	220	1,105
1,500	6,864 / 1,834 / 1,229	9.76	28.23	220	1,308

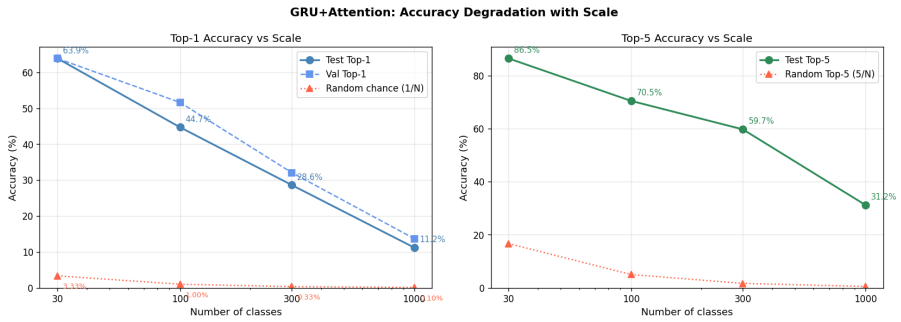


Figure 6.4: Top-1 accuracy as a function of vocabulary size on WLASL. Source: self-created from experiment outputs.

The relationship between vocabulary size and Top-1 accuracy is monotonically decreasing across the entire range and roughly linear in $\log(\text{num_classes})$ between 30 and 1,000 classes. The shape

of the curve has a clear interpretation: the model is not breaking. WLASL2000 has a median of approximately ten samples per class, which is below the regime where a discriminative neural sequence model can separate fine-grained gestural categories without external pretraining or stronger augmentation. The bottleneck is the per-class data budget, not the architecture.

A weaker secondary observation, restricted to the runs above 700 classes, is that the training curves of the larger subsets show greater sensitivity to early-stopping triggers; runs at 900, 1,200, and 1,500 classes saturate the maximum-epoch budget without an unambiguous improvement, while runs below 700 classes consistently early-stop earlier. This is the practical edge of the data-starvation regime: not a discrete failure point, but a regime where additional training time stops paying back.

The data-starvation interpretation is corroborated by the parallel-coordinates view in Figure 6.5, which projects every scaling run onto five normalised axes (mean videos per class, final train accuracy, validation Top-1, test Top-1, and test Top-5). The signature of the regime is the sharp drop between train accuracy and validation Top-1: at 30 classes the line stays high across all axes (mean 11.07 videos per class, train accuracy 99.00 %, val Top-1 63.85 %), whereas at 1,500 classes the same line collapses from 74.77 % train accuracy to 11.67 % validation Top-1 (4.58 videos per class, according to the scaling summary). The lines do not converge gradually, because the train-side stays comparatively flat while every validation- and test-side axis crashes. This is the textbook fingerprint of overfitting driven by sample scarcity rather than by an inadequate inductive bias.

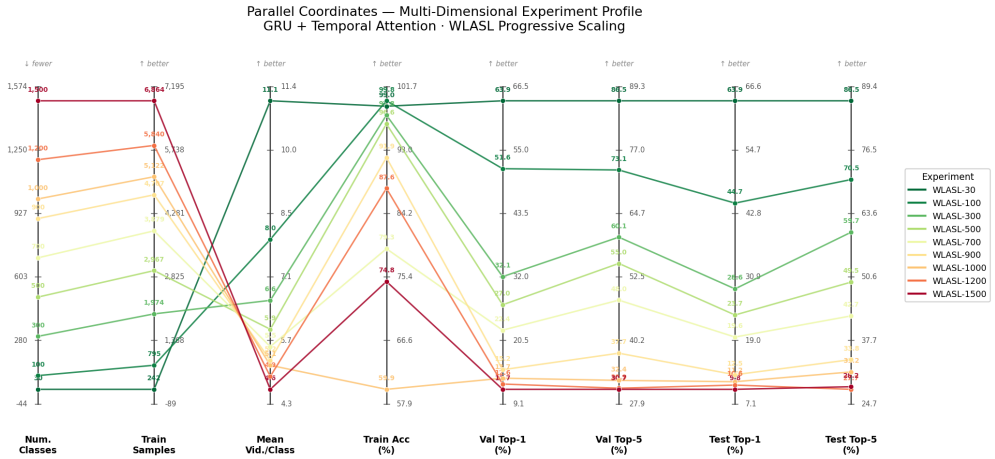


Figure 6.5: Parallel-coordinates view of the GRU+Attention scaling line on WLASL. Each line is one experiment, coloured from green (few classes) to red (many). The widening train–val gap is the hallmark of data starvation. Source: self-created from experiment outputs.

6.8 Limitations and Threats to Validity

The experiments above support several claims and undermine several others. The genuine limitations fall into six categories, addressed in turn.

The MediaPipe detection rate on WLASL is approximately 69 % at the frame level. Clips with motion blur, partially occluded hands, or hands moving outside the frame produce silent failures that the linear-interpolation step mitigates but does not erase. The interpolated frames carry no new

information, so any clip with sustained low detection rate effectively provides fewer informative frames than its nominal length suggests. The same caveat applies less forcefully to MS-ASL50, where the average detection rate is $83.9\,\%$.

The WLASL30 test split is small (34 samples). Even at three seeds, the standard deviation in Top-1 is bounded below by $\approx$2.94 pp because that is the resolution of one misclassification. The 58.82 ± 2.94 figure should be read as a range that any reasonable run will fall into, not as a precise estimate of the true generalisation accuracy. The 5-Fold CV mean of 63.85 partially mitigates this by averaging over five different test partitions, but the absolute resolution per fold is similar. This caveat is the reason MS-ASL50, with its 236-sample test split, is a more useful benchmark for fine-grained per-class analysis.

Beyond 30 classes, every reported result is single-seed. Differences between adjacent vocabulary sizes in Section 6.7 (e.g., 23.71 at 500 vs. 19.57 at 700) sit within a plausible seed-to-seed noise band, especially because the test sets at those sizes are large enough to make the per-sample resolution finer than at WLASL30. The downward trend itself is robust; the exact slope is not.

YouTube attrition on MS-ASL is $34.1\,\%$ of source clips. The MS-ASL50 subset reproducible from current YouTube state is therefore not identical to the one reproducible six months ago and will not be identical to the one reproducible six months from now. Any direct numerical comparison with results published against the original MS-ASL release [3] must be read with this caveat.

Only one architecture (GRU+Attention) was evaluated cross-dataset. A Transformer run on MS-ASL50 would directly test whether the Transformer's underperformance in Section 6.3 reflects the WLASL30 data regime or a deeper limitation of the architecture choice. That experiment is identified as a natural extension but was not executed within the scope of this thesis.

The augmentation pipeline was kept fixed throughout the chapter: rotation, scaling, translation, Gaussian noise, and horizontal flip. Horizontal flip in particular conflates left-handed and right-handed signers, which is acceptable for a small vocabulary one-hand pipeline but would need re-examination for two-hand signs and for downstream studies of signer-handedness bias. Class-weighted loss and minority-class oversampling were also not applied, despite the per-class disparities documented in Sections 6.5 and 6.6. Both are explicit future-work items.

Part II: Qualitative Evaluation

6.9 Qualitative Validation Protocol (In Progress)

The metrics reported in Part I measure recognition accuracy on fixed test sets under controlled landmark extraction. They do not, on their own, measure the usability of the mobile prototype when a real user tries to operate it. Closing that gap requires a structured protocol with human participants, the design of which is the subject of this section. The protocol is fixed; data collection and analysis are pending and will appear in a complementary report rather than in this thesis. The five subsections below cover the motivation and research questions, the participant population and recruitment criteria, the per-session task design, the instrumentation that records each session, and an analysis plan that defines how the qualitative session data will be reduced to quantitative metrics for that report.

6.9.1 Motivation and Research Questions

The qualitative protocol is necessary in addition to the quantitative experiments, not in place of them. Part I demonstrates that the model learns a recognisable representation of the MS-ASL50 vocabulary; the qualitative protocol is the first evidence that the surrounding system serves a user. Three properties of the prototype are invisible to Part I and motivate the protocol explicitly. The quantitative experiments consume clips that were segmented offline and processed by MediaPipe on the workstation, whereas the prototype consumes clips captured from a phone camera under uncontrolled lighting, framing, and execution speed. The Top-5 mechanism that this chapter relies on as an accountability fallback is only useful if participants actually consult it, and that behaviour is not observable from server-side metrics alone. Finally, the latency target stated as NFR-001 is meaningful only when measured end-to-end against real network and capture conditions, not against an isolated server benchmark.

The protocol is therefore designed to answer four research questions that Part I does not address. *RQ1:* Does the end-to-end latency of the prototype meet the $\leq$ 2-second target stated in NFR-001 under live use, as opposed to under isolated server benchmarking? *RQ2:* When the Top-1 prediction is incorrect, do participants consult the Top-5 fallback, or do they ignore it and treat the prediction as a single forced choice? *RQ3:* Which capture frictions, namely lighting, framing, distance from camera, and execution speed, dominate the user-perceived error rate, relative to the residual model error documented in Part I? *RQ4:* Does the Top-1 figure reported in Section 6.6 (70.20 % on MS-ASL50) overestimate or underestimate the accuracy a real user perceives in the field?

6.9.2 Participants and Recruitment

The target sample size is 10 to 25 participants. The lower bound is set so that variance across individuals does not dominate any aggregate observation; the upper bound is set so that the protocol stays within the regime of heuristic usability evaluation rather than crossing into a formal statistical study, which would require demographic stratification and controls that are out of scope here. The sample is not statistically representative of the deaf or hard-of-hearing population; it is an instrument for surfacing usability obstacles, not a census of the user base.

Inclusion criteria are an explicit interest in accessibility or assistive technology, or non-native ASL fluency motivated by an interest in learning the language. Native and fluent ASL signers are excluded from this round because they are not the population the prototype targets and because their natural execution speed and precision would bias the capture-friction observations toward zero. Recruitment is conducted through the author's personal and academic network, university accessibility-focused groups, and equivalent informal channels. Each participant signs a consent form acknowledging that the session captures server-side request logs, Likert responses, and free-text comments; that captured video frames are not persisted beyond the session; and that aggregated results may appear in the complementary report.

6.9.3 Task Design and Stimuli

Each session is structured around a fixed subset of $N = 10$ glosses drawn from the MS-ASL50 vocabulary. The choice of N is a compromise between three constraints: it must cover both ends of the bimodal per-class distribution documented in Section 6.6, it must fit a session of approximately 15 to 25 minutes without inducing fatigue, and it must include enough trials per participant for the per-trial outcome series to carry information. The 10 glosses are selected by stratification, not by

sampling: five are drawn from the classes that reached perfect recall on the seed-2026 test split (for example *teacher*, *no*, *tired*), and five are drawn from the classes whose recall fell below 0.50 (for example *here*, *man*, *woman*, *need*). The 1:1 stratification ensures that each session exposes both the best-case and the worst-case behaviour of the model.

The per-gloss procedure is the same for every participant. The participant first sees a short reference clip of a correctly executed sign, taken from a ground-truth video that was not part of the training set. The participant then performs the sign in front of the phone camera, and the prototype returns a Top-5 prediction with associated confidence scores. The participant marks three outcomes per trial: whether the Top-1 matches the target gloss, whether the Top-5 contains the target gloss, and an optional free-text comment. Each gloss is performed twice. The first attempt measures the user's learning curve with respect to the application itself; the second measures the steady-state accuracy after one round of feedback. The order of glosses is randomised per participant to neutralise ordering and fatigue effects.

6.9.4 Instrumentation: Server Logs and Google Form

Each session produces four streams of data. The first is server-side request data: every `POST /predict` call carries a timestamp, the measured end-to-end latency, the Top-K predictions with confidence scores, and an anonymised `participant_id`. The current FastAPI server documented in Section 5.1 exposes a `save_debug` hook that already persists the response payload to disk; for the qualitative sessions a thin request-logging middleware records the additional fields above into a structured JSONL file per session. The second stream is a post-session Google Form, organised in two blocks: a Likert block (1–5) covering perceived accuracy, frustration, intention to use the system in the future, and clarity of the Top-5 feedback, followed by a free-text block for capture-related obstacles. The third stream is the per-trial outcome that the participant marks during the session itself: the two binary fields *Top-1 matched the target* and *Top-5 contained the target*. These two fields, joined with the server logs, produce the per-trial time series on which the analysis plan in Section 6.9.5 operates. The fourth stream is observational: the author records, by hand and per gloss, retries and abandonment events. Detecting that a participant repeated a sign without success and moved on is a proxy for friction that the post-session Form does not capture as cleanly.

6.9.5 Analysis Plan: Quantifying the Qualitative

The reduction from session data to quantitative metrics is fixed in advance, in the spirit of preregistration, so that the qualitative chapter cannot be tuned to flatter the prototype after the fact. Six metric families will be derived once collection completes.

The first is the *user-perceived Top-1 accuracy*, defined as the number of trials whose Top-1 matches the target divided by the total number of trials. It is directly comparable to the 70.20 % reported in Section 6.6, with the caveat that trials are now over user-captured clips rather than over the MS-ASL50 test split. The second is the *user-perceived Top-5 accuracy*, defined analogously and comparable to the 90.54 %. The third is the *Top-5 utility rate*, defined as the proportion of Top-1-incorrect trials in which the participant reports having consulted the Top-5; this is the operational definition of RQ2 and is the only one that requires a self-report field. The fourth is the *median end-to-end latency* per trial, computed from the timestamps in the server log; the comparison against the 2-second NFR-001 target answers RQ1. The fifth is a *capture-friction frustration index*, defined as the per-category count of free-text comments classified by hand under the labels {*lighting*, *framing*, *distance*, *execution speed*, *other*}; the resulting distribution is the operational decomposition of RQ3. The sixth is

a per-item *Likert mean with* $95\,\%$ *bootstrap confidence interval* (10 000 resamples), used because at N between 10 and 25 the parametric normal-based interval is unreliable.

A single hypothesis test is pre-declared. To answer RQ4, the user-perceived Top-1 accuracy will be compared against the $70.20\,\%$ reference using a one-sample z-test of proportions; the conclusion will be reported as direction (*overestimates* or *underestimates*) and effect size, not as a binary significance verdict, given the modest sample size.

These analyses are pre-registered in the protocol and will be applied unchanged to the data once collected. The qualitative chapter does not anticipate any specific outcome.

Chapter 7

Ethical Implications and Social Impact

7.1 AI Based Systems Ethical Impact

An ethical analysis in Artificial Intelligence examines the implications, concerns and challenges associated with the development and deployment of these systems. As AI integrates into increasingly critical applications, technical evaluation must be paired with an explicit ethics review. This review rests on four general pillars: **Human Rights and Fairness** (preventing bias and discrimination), **Accountability** (clarifying who is responsible for the system's decisions), **Privacy and Security** (protecting user data and confidentiality), and **Transparency** (avoiding opaque black-box behaviour).

7.2 Ethical Evaluation Framework: Google's AI Principles

The framework adopted for this evaluation is **Google's AI Principles** [15], originally published in 2018 and refined annually; the primary reference for this analysis is the **2023 Progress Update** [16]. The framework was chosen because its seven principles are formulated as concrete objectives that map cleanly onto the design decisions of a research project, and because Google's annual progress reports demonstrate how the principles apply to systems that share characteristics with this work (computer vision and on-device inference). The seven principles are:

1. **Be socially beneficial:** address a real societal need and do so effectively.
2. **Avoid creating or reinforcing unfair bias:** ensure fair and inclusive treatment.
3. **Be built and tested for safety:** prioritise safety in design, validation, and operation.
4. **Be accountable to people:** provide transparency and user control.
5. **Incorporate privacy design principles:** protect user data by design.
6. **Uphold high standards of scientific excellence:** follow rigorous methodology.

7. **Be made available for uses that accord with these principles:** prevent misuse and align deployment with the framework.

The remainder of the chapter evaluates the project against each principle through quantitative metrics. Each metric is scored on a simplified {0, 5, 10} scale (not addressed, partially addressed, fully addressed). Some metrics derive from experimental results; others reflect design decisions or documentation practices. The summary table at the end of the chapter aggregates the metrics into per-principle averages and a global score.

7.3 Project Analysis Against Google's AI Principles

7.3.1 Principle 1: Be Socially Beneficial

Sign language is the primary means of communication for millions of deaf and hard-of-hearing individuals, yet a substantial communication barrier persists with the hearing population. An automatic ASL recognition system can act as a bridge in educational, medical, and everyday contexts.

- **M1.1: Real societal need.** The World Health Organization estimates over 430 million people worldwide have disabling hearing loss; the project targets a clear accessibility problem. *Score: 10.*
- **M1.2: Effectiveness of the solution.** The champion model achieves **63.85% Top-1 / 86.46% Top-5 (5-fold cross-validation, WLASL30)**, reproduced under the modular pipeline as 58.82% $\pm$ 2.94% Top-1 across three seeds, and extends to **70.20% $\pm$ 2.89% Top-1 / 90.54% $\pm$ 0.89% Top-5** on MS-ASL50. Accuracy degrades at scale (44.70% at 100 classes, 28.61% at 300, 11.19% at 1000), so the system is a research prototype rather than a deployment-ready translator. *Score: 5.*
- **M1.3: Accessibility of the technology.** Built on open-source tools (Python, PyTorch, MediaPipe) and public datasets (WLASL, MS-ASL), with the methodology documented in this thesis. The recognition pipeline was designed with mobile-first deployment in mind. *Score: 10.*

7.3.2 Principle 2: Avoid Creating or Reinforcing Unfair Bias

Sign language recognition is particularly bias-sensitive: signers vary in hand size, skin tone, signing speed, and style. Bias arises both at the perception layer (hand detection / landmark extraction operating on raw images) and at the recognition layer.

- **M2.1: Dataset diversity (word recognition).** WLASL aggregates videos from multiple online sources, providing implicit signer variation, but lacks demographic annotations (skin tone, age, hand morphology), so diversity cannot be quantified formally. *Score: 5.*
- **M2.2: Dataset diversity (hand detection and letter classification).** Hand detection used EgoHands ($\sim$4,800 egocentric images) and the Kaggle Hand Dataset ($\sim$4,200 third-person images). Letter classification used Sign Language MNIST (34,627 grayscale images) and the ASL Alphabet Dataset ($\sim$87,000 RGB images). None provide demographic metadata, so potential biases cannot be ruled out. *Score: 5.*
- **M2.3: Input representation fairness.** The word-recognition model consumes MediaPipe landmarks (21 keypoints $\times$ 3 coordinates) instead of raw pixels. Operating on geometric joint

positions rather than visual appearance reduces sensitivity to skin colour, lighting, and background, mitigating appearance-based bias at the recognition stage (though not at the upstream landmark extractor). *Score: 10.*

- **M2.4: Per-class performance analysis.** Per-class metrics were logged across all experiments, surfacing wide accuracy gaps (e.g. "what" 8.3%, "cool" 12.5%, vs. "orange" 100%, "table" 100%). The disparities are mostly explained by visual similarity between signs and small per-class sample counts rather than by demographic factors. Class-weighted loss or oversampling were not applied in this iteration but are facilitated by the existing per-class logging. *Score: 5.*

7.3.3 Principle 3: Be Built and Tested for Safety

In this context, safety means producing reliable predictions and avoiding misleading outputs that could cause misunderstandings in sensitive domains.

- **M3.1: Robust validation methodology.** Evaluation uses Stratified 5-Fold Cross-Validation (mean Top-1 **63.85% ± 4.50%**, mean Top-5 **86.46% ± 2.04%**), corroborated by a 3-seed multi-seed sweep on the same dataset (58.82% ± 2.94% Top-1) under the refactored pipeline. Test-Time Augmentation was deliberately avoided because it inflates reported accuracy without reflecting true generalisation. *Score: 10.*
- **M3.2: Scalability testing.** The model was retrained on progressively larger WLASL subsets (30 / 100 / 300 / 500 / 700 / 900 / 1000 / 1200 / 1500 classes), documenting accuracy degradation and an instability point above 700 classes. This stress-test establishes the boundaries within which the system can be considered reliable. *Score: 10.*
- **M3.3: Hand detection reliability.** The YOLOv8n hand detector reached approximately 98% mAP50 on the combined test set, validated across both source datasets. A reliable detector is a prerequisite for the recognition pipeline. *Score: 10.*
- **M3.4: Failure mode documentation.** Per-class confusion matrices identify systematically confused word pairs, but a confidence-threshold mechanism (so the system can abstain when uncertain) is not yet implemented and is needed before any production deployment. *Score: 5.*

7.3.4 Principle 4: Be Accountable to People

Accountability requires that the system is interpretable to its users and gives them meaningful control over its outputs.

- **M4.1: Model interpretability.** The Temporal Attention layer produces per-frame weights that can be inspected to understand which frames drove a prediction. This provides a degree of post-hoc interpretability that fully opaque architectures lack. *Score: 10.*
- **M4.2: Top-K predictions.** The system exposes Top-5 alongside Top-1. With Top-5 accuracy of 86.46% on WLASL30 (and 90.54% on MS-ASL50), the correct answer is almost always in the shortlist, allowing the user to make the final call rather than accepting a single forced prediction. *Score: 10.*
- **M4.3: Reproducibility.** All experiments are driven by versioned YAML configs and produce manifests with git commit, hyperparameters, and metrics. The main caveat is that some

WLASL videos rely on YouTube URLs that may decay over time; this is also an ethical concern about data provenance, since the original signers did not explicitly consent to use of their videos for ML research. *Score: 5.*

7.3.5 Principle 5: Incorporate Privacy Design Principles

Privacy applies to three distinct stages of the project: training the hand detector on real photographs, training the word recognition model on landmark abstractions, and serving inference from a mobile client.

- **M5.1: Data minimisation through landmarks (training).** Each video frame is reduced to 63 numerical values that discard the original visual content. Once landmarks are extracted, the original video is no longer needed and the processed data cannot reconstruct the signer's face, body, or background. *Score: 10.*
- **M5.2: Privacy in the hand detection phase.** Detector training relied on real photographs of identifiable individuals (EgoHands, Kaggle Hand). Although these datasets are publicly available for research, no independent privacy audit was conducted on their provenance, and the images were used as-is during training. *Score: 5.*
- **M5.3: Privacy in the inference pipeline.** The current FastAPI server receives full RGB frames (416×416) from the mobile app, processes them, and persists raw frames and cropped regions to disk without encryption or automatic expiration. This is the weakest privacy point. A natural mitigation is to move landmark extraction to the client device so that only the abstract 63-dimensional vector ever leaves the phone. *Score: 0.*
- **M5.4: K-anonymity considerations.** WLASL stores no identifying metadata about signers, and the landmark representation acts as a generalisation: different individuals performing the same sign yield similar feature vectors [17]. A formal k-anonymity algorithm was not applied, but the pipeline achieves a comparable effect for the recognition stage. The protection does not extend to the detector training data nor to the server inference path. *Score: 5.*

7.3.6 Principle 6: Uphold High Standards of Scientific Excellence

- **M6.1: Systematic model comparison (word recognition).** Multiple architectures were trained under matched conditions: CNN+LSTM (pixel-based), LSTM (landmark-based), plain GRU, GRU with enriched features, GRU + Temporal Attention, and a Transformer variant. *Score: 10.*
- **M6.2: Systematic model comparison (hand detection).** The hand detection stage compared YOLOv8n, SSDLite-MobileNetV3, and EfficientDet-D0 on the same combined dataset using standard detection metrics (mAP50, mAP50-95, latency). *Score: 10.*
- **M6.3: Statistical validation.** Stratified 5-Fold Cross-Validation reports mean and standard deviation across folds ($63.85\% \pm 4.50\%$ Top-1), and the modular pipeline contributes additional multi-seed runs on WLASL30 and MS-ASL50 to bound model-init variance. Pearson correlation analysis quantifies the relationship between vocabulary size and accuracy. *Score: 10.*
- **M6.4: Literature grounding.** The State of the Art chapter situates each architectural choice (CNN, RNN, LSTM, GRU, Attention, MediaPipe) in the established literature. *Score: 10.*

- **M6.5: Honest reporting of limitations.** The thesis explicitly documents accuracy degradation at scale, the ∼69% MediaPipe detection rate on WLASL, and the instability observed above 700 classes. Negative results are reported alongside positive findings. *Score: 10.*

7.3.7 Principle 7: Be Made Available for Uses That Accord with These Principles

- **M7.1: Intended use clarity.** The thesis frames the system as a research prototype for ASL word recognition, with assistive technology as the target use case. Scope and limitations are explicit. *Score: 10.*
- **M7.2: Misuse potential assessment.** Landmark vectors carry very limited misuse potential (no surveillance or facial recognition is recoverable from 63 floats per frame). The detector and the inference server, however, do process raw images and currently lack access controls and retention policies; this gap should be closed before any deployment beyond the research context. *Score: 5.*
- **M7.3: Open research contribution.** The methodology and code are documented in the thesis and in the repository. Trained models and processed datasets have not yet been published in a public artifact registry. *Score: 5.*

7.4 Impact Scale of Ethical Aspects

Table 7.1 consolidates all metrics, their individual scores, and the per-principle average.

Principle	ID	Metric	Score	Avg.
P1: Socially Beneficial	M1.1	Addresses a real societal need	10	**8.3**
	M1.2	Effectiveness of the solution	5	
	M1.3	Accessibility of the technology	10	
P2: Avoid Unfair Bias	M2.1	Dataset diversity (word recognition)	5	**6.3**
	M2.2	Dataset diversity (detection & letters)	5	
	M2.3	Input representation fairness	10	
	M2.4	Per-class performance analysis	5	
P3: Safety	M3.1	Robust validation methodology	10	**8.8**
	M3.2	Scalability testing	10	
	M3.3	Hand detection reliability	10	
	M3.4	Failure mode documentation	5	
P4: Accountability	M4.1	Model interpretability	10	**8.3**
	M4.2	Top-K predictions	10	
	M4.3	Reproducibility	5	
P5: Privacy	M5.1	Data minimisation (landmark training)	10	**5.0**
	M5.2	Privacy in detection phase	5	
	M5.3	Privacy in inference pipeline	0	
	M5.4	K-anonymity considerations	5	
P6: Scientific Excellence	M6.1	Model comparison (word recognition)	10	**10.0**
	M6.2	Model comparison (hand detection)	10	
	M6.3	Statistical validation	10	
	M6.4	Literature grounding	10	
	M6.5	Honest reporting of limitations	10	
P7: Principled Availability	M7.1	Intended use clarity	10	**6.7**
	M7.2	Misuse potential assessment	5	
	M7.3	Open research contribution	5	
		Overall Average Score		**7.6**

Table 7.1: Ethical evaluation metrics based on Google's AI Principles [16]

7.5 Conclusions of the Ethical Analysis

The ethical evaluation yields an overall score of **7.6 out of 10**. The strongest pillars are scientific excellence (10.0/10) and safety (8.8/10); the weakest is privacy (5.0/10), driven entirely by the inference pipeline.

One architectural decision, the use of MediaPipe hand landmarks as input, propagates positive effects across multiple principles: it reduces appearance-based bias (P2), it implements data minimisation that protects identity by construction (P5), and it limits misuse potential because abstract landmarks cannot be repurposed for surveillance (P7). The same advantage, however, does not extend to the upstream stages that operate on raw images. The current FastAPI server receives, processes, and persists full RGB frames without encryption or automatic deletion, and the hand detector was trained on photographs of identifiable individuals. The most actionable mitigation is to relocate landmark extraction to the client device, so that only the 63-dimensional vector ever leaves the phone.

The intermediate scores in bias avoidance (6.3/10) and principled availability (6.7/10) reflect missing formal processes rather than poor design choices: no demographic audit of signers was conducted, the auxiliary detection datasets lack diversity metadata, and the server prototype has no access controls or retention policies. Honest reporting of these gaps, including the privacy weakness of the

inference pipeline identified in this chapter and the documented accuracy collapse at large vocabularies, is itself an ethical practice aligned with Principle 6 and is a prerequisite for any responsible deployment of the system.

Chapter 8

My Journey at UFV

Chapter 9

Conclusiones

9.1 General Conclusions

9.2 Conclusions regarding Project Objectives

9.3 Next Steps

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